



FACULTAD DE CIENCIAS

SOBRE LA ACCIÓN DEL GRUPO DE AUTOMORFISMOS DE UNA SUPERFICIE DE RIEMANN COMPACTA SOBRE EL CONJUNTO DE SUS SÍSTOLES

Tesis

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Camila Guajardo Vásquez ha vivido siempre en el mismo edificio de Santiago Centro, Santiago, Chile. De pequeña soñaba con dedicarse al atletismo y luego a la neurología, pero el diagnóstico de una artritis crónica a los 15 años le hizo cuestionarse qué quería para su futuro.

En medio de la incertidumbre tuvo a un profesor, Sergio Santelices (QEPD), que confió en su talento para resolver problemas de olimpiadas matemáticas. Esas olimpiadas fueron el comienzo de una intensa atracción por la geometría y un nuevo sueño: convertirse en investigadora.

Se graduó como Licenciada en Matemática por la Pontificia Universidad Católica de Chile en 2023 y decidió continuar su camino de geómetra en la Universidad de Chile. Al término de esta tesis tiene 24 años.

Sigue interesándole la neurología y descubrió el gusto por la psicología, la botánica, y la literatura (existencialista y asiática de no ficción). Es una ilustradora botánica amateur, una activa visitante del cerro San Cristóbal y una empedernida lectora de transporte público.

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Resumen

En Geometría Riemanniana, la noción de simetría la encarnan las *isometrías*, mientras que en Geometría Compleja lo hacen los *automorfismos*. ¿Cómo interactúan las estructuras Riemanniana y compleja en una superficie de Riemann compacta dotada de una métrica Riemanniana? Ciertas geodésicas cerradas simples nos dan una respuesta: las *sístoles* de la superficie. Las conjeturas sugieren que tener grupos de automorfismos de orden grande implican tener mayor *kissing number* (número de sístoles) y que sean de mayor largo, y viceversa.

En esta tesis desarrollamos e implementamos un algoritmo para encontrar sístoles a partir de la acción de un grupo abstracto. Más aún, obtuvimos cotas explícitas en familias de superficies dotadas de una acción de grupo no trivial: inferiores para el *kissing number* y superiores para el largo de las sístoles.

Abstract

In Riemannian Geometry, the notion of symmetry is captured by the *isometries*, while in Complex Geometry, it corresponds to the *automorphisms*. How do the Riemannian and complex structures interact on a compact Riemann surface endowed with a Riemannian metric? A certain kind of simple closed geodesics tells us an answer: the *systoles* of the surface. The conjectures suggest that larger automorphism groups imply larger kissing numbers (the number of systoles) and longer systoles, and vice versa.

In this thesis, we develop and implement an algorithm to find systoles starting with an action of an abstract group. Moreover, we have obtained explicit lower bounds on the kissing number and upper bounds on its lengths for families of surfaces with non-trivial group actions.

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Chapter 1

Introduction

Let S be a compact Riemann surface of genus $h \geq 2$. By the Uniformization theorem, such a surface can be realized as a quotient of the unit disc $\mathbb{D}$ by a torsion-free Fuchsian group Γ . The unit disc $\mathbb{D}$ in $\mathbb{R}^2$ is a Riemannian manifold of dimension 2 with the following Riemannian metric, which is given in complex local coordinate $z = x + iy$ as

$$g_z = \frac{4}{(1 - |z|^2)^2} dz d\bar{z}. \quad (1.1)$$

The isometry group of $(\mathbb{D}, g)$ contains the analytic automorphisms of $\mathbb{D}$, that is, the group

$$\text{Aut}(\mathbb{D}) = \left\{ z \mapsto e^{i\theta} \frac{z - a}{1 - \bar{a}z} : \theta \in [0, 2\pi], a \in \mathbb{D} \right\}, \quad (1.2)$$

and the geodesics on $\mathbb{D}$ are arcs of Euclidean circles which are orthogonal to the boundary of $\mathbb{D}$, or line segments passing through the origin (see [Anderson, 2006]).

The metric on $\mathbb{D}$ induces a Riemannian metric on S such that the covering map $\pi: \mathbb{D} \rightarrow \mathbb{D}/\Gamma$ is a local isometry. Thus, the surface S inherits a Riemannian structure from $\mathbb{D}$, and the geodesics on S are the projections of geodesics lying in a fundamental polygon for the action of Γ on $\mathbb{D}$.

Let P be such a fundamental polygon. The sides of P are identified in pairs by the action of elements of Γ . For each side e in ∂P , there is a unique side e' and an element $f \in \Gamma$ such that $f(e) = e'$. A *closed geodesic* on S is a geodesic on $\mathbb{D}$ contained in P whose endpoints coincide or they are boundary points identified by the action of Γ (see figure 1.1).

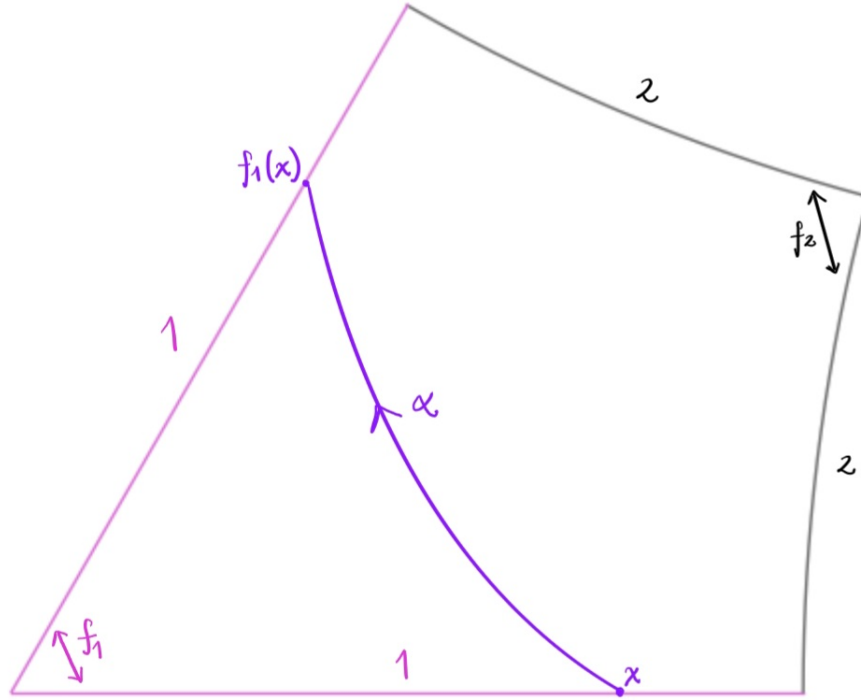


Figure 1.1: Closed geodesic on a fundamental polygon. The sides labeled with the same number are identified.

Each homotopy class of simple curves (that is, with no self-intersections) with fixed endpoints has a geodesic representative ([Jost, 2017, Theorem 1.6.1]). For a simple closed curve α on S , its free homotopy class $[\alpha]$ contains simple closed geodesic homotopic to α . Among these, some realize the minimum length in $[\alpha]$; denote such a closed geodesic as $s([\alpha])$.

Definition 1.1 (Systoles). A *systole* of S is a simple, closed geodesic $s([\alpha])$ such that for every closed curve β in S , not freely homotopic to α , the length of $s([\alpha])$ is least of equal to the length of $s([\beta])$.

A systole is a non-homotopically trivial simple closed geodesic of minimum length in S , associated to a unique free homotopy class. The figure 1.2 illustrates this concept.

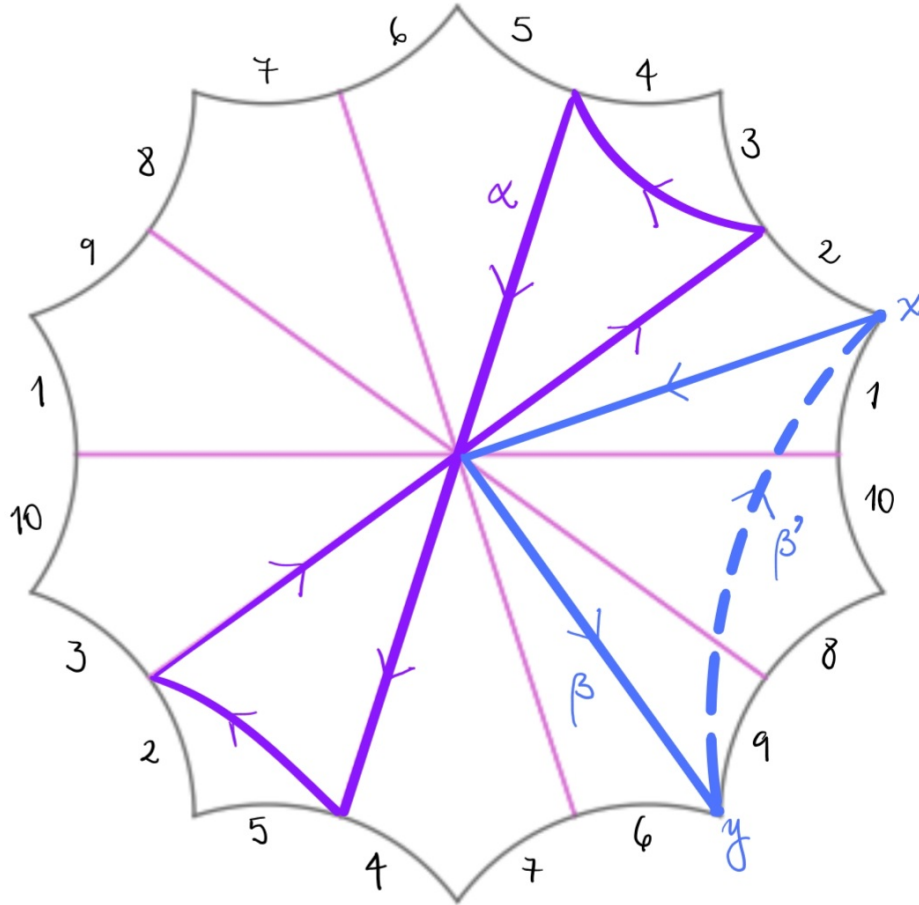


Figure 1.2: Some closed curves on a surface of genus two. The purple curve α is piecewise geodesic, but not simple. The blue curve β is a simple closed piecewise geodesic, homotopic to the systole β' .

There are two isometric invariants associated to the systoles of S :

Definition 1.2 (Systolic length and Kissing number). The length of a systole is called the *systolic length* of S , and is denoted by $\text{sys}(S)$. The number of systoles is called the *kissing number* of S , and is denoted by $\text{Kiss}(S)$.

Computing bounds for these invariants is generally difficult. In genus two, the biggest kissing number and systolic length are reached by the *Bolza Surface* (see [Schmutz, 1993, Theorem 5.2]). For higher genus, the con-

tures state that there are genus-dependent upper bounds for both quantities. More precisely, there are universal constants $K, L > 0$ that

$$\max_{S \in \mathcal{M}_h} \text{sys}(S) \leq \frac{4}{3} \log(h) + K, \quad (1.3)$$

$$\max_{S \in \mathcal{M}_h} \text{Kiss}(S) \leq Lh^{4/3}, \quad (1.4)$$

with $\mathcal{M}_h$ the moduli space of compact Riemann surfaces of genus h .

The excellent survey [Schmutz-Schaller, 1998] collects the main conjectures, results and perspectives of the geometry of Riemann surfaces based on the study of systoles. More on the kissing number can be found in [Parlier, 2013].

Our objective is to compute lower bounds on the kissing number and upper bounds on the systolic length in the case of families of surfaces with a non-trivial *group action*.

Definition 1.3 (Group action on Riemann surfaces). Let G be an abstract finite group. A *group action* of G on S is an injective homomorphism $\varphi: G \rightarrow \text{Aut}(S)$.

Suppose that S admits a group action $\varphi: G \rightarrow \text{Aut}(S)$. Then, the quotient surface S/G can be endowed with a complex structure such that the quotient map $\pi_G: S \rightarrow S/G$ is a branched covering map. The branched values of π_G correspond to the points $q \in S/G$ such that its fiber $\pi_G^{-1}(q)$ has less elements than $|G|$. Each point $p \in \pi_G^{-1}(q)$ has a non trivial stabilizer subgroup of order m in $\varphi(G)$. It is well known that such stabilizers must be cyclic subgroups and points in the same fiber have conjugated stabilizers, so $\pi_G^{-1}(q)$ consist of $|G|/m$ elements. See [Carvacho Bustamante, 2010, Section 1.2] for a good review.

Thus, if $q_1, \dots, q_r$ are the branched values of π_G and $m_1, \dots, m_r$ are the orders of the corresponding stabilizers of points in their fibers, we say that G *acts with signature* $(\gamma; m_1, \dots, m_r)$ on S . Here γ is the genus of the quotient surface S/G .

In this setting, the Riemann-Hurwitz formula states that

$$2h - 2 = |G| \left(2\gamma - 2 + \sum_{j=1}^r \left(1 - \frac{1}{m_j} \right) \right). \quad (1.5)$$

Let's look at the Fuchsian groups context. Let Δ be a Fuchsian group that uniformizes the quotient surface S/G . The group action $\varphi: G \rightarrow$

$\text{Aut}(S)$ is equivalent to a *surface-kernel epimorphism* $\theta: \Delta \rightarrow G$, that is, a group epimorphism whose kernel is isomorphic to Γ , the Fuchsian group that uniformizes S (see [Carvacho Bustamante, 2010, Section 1.3]). If G acts with signature $\sigma = (\gamma; m_1, \dots, m_r)$ on S , the group Δ has presentation

$$\Delta = \left\langle a_1, b_1, \dots, a_\gamma, b_\gamma, c_1, \dots, c_r \mid c_1^{m_1} = \dots = c_r^{m_r} = \prod_{i=1}^{\gamma} [a_i, b_i] \cdot \prod_{j=1}^r c_j = 1 \right\rangle. \quad (1.6)$$

Thus, the element $\theta(c_j)$ must have order m_j in G , for each $j = 1, \dots, r$. The list $v = (\theta(a_1), \theta(b_1), \dots, \theta(a_\gamma), \theta(b_\gamma); \theta(c_1), \dots, \theta(c_r))$ is called a *generating vector* for G for the signature σ .

Riemann's existence theorem (see [Broughton, 1991, Proposition 2.1]) gives a criterion to determine when a finite group acts with a given signature on a Riemann surface: A group G acts with signature $\sigma = (\gamma; m_1, \dots, m_r)$ on a Riemann surface of genus h if and only if the Riemann-Hurwitz formula holds and G has a generating vector v for the signature σ .

An automorphism F of $\mathbb{D}$ descends to an automorphism f of S if and only if $F \circ \Gamma \circ F^{-1} = \Gamma$. As the automorphisms of $\mathbb{D}$ are isometries, the group $\text{Aut}(S)$ acts by isometries on S . Therefore, an automorphism f of S sends geodesics to geodesics, and in particular, systoles to systoles.

Our strategy for computing lower bounds on the kissing number starts with a group action. In [Broughton, 1991, Table 4, p.252] are listed the possible group actions on genus two, so we can choose the ones with bigger groups. Then, we construct a fundamental polygon P for a surface S admitting such a group action, based on the signature and a generating vector. We follow the methods developed in [Behn et al., 2013]. Finally, we look for closed geodesics in P that minimize the length among their homology class. These closed geodesics are candidates to be systoles.

Remark 1.4 (Philosophy of this thesis). *The length of these curves is an upper bound for the systolic length, and the sizes of their orbits under the action of the abstract group are lower bounds for the kissing number.*

This thesis is organized as follows. In chapter 2, we develop in detail our strategy in the particular case of the Bolza surface. We follow the methods, algorithms and conventions of [Behn et al., 2013]. In chapter 3, we focus on the other genus two surfaces with a non-trivial group action. We adapt the steps from the previous chapter to the corresponding signatures and generating vectors, and we finish with a comparative table of the systolic bounds we found. Finally, chapter 4 is an appendix containing the Sage code

that we implement and use along this work, which is posted in the GitHub repository available in <https://github.com/Cami-GV/Master-thesis.git>

For more details on Riemannian manifolds, we refer the reader to

- the book *Introduction to Smooth manifolds* [Lee, 2003],
- the article *Space of Riemannian metrics* [Bustamante and Kordaß, 2017] (for a more direct approach to Riemannian metrics),
- and the book *Introduction to Riemannian manifolds* [Lee, 2018].

For hyperbolic geometry, we refer to

- the book *Hyperbolic Geometry* [Anderson, 2006],
- and the book *The geometry of discrete groups* [Beardon, 2012] for the Fuchsian groups approach.

Finally, for group actions on Riemann surfaces, we refer to

- the article *Classifying finite group actions on surfaces of low genus* [Broughton, 1991],
- the survey *A Fruitful Interaction Between Algebra, Geometry, and Topology: Varieties Through the Lens of Group Actions* [Rodríguez and Rojas, 2024] (for a broader perspective on moduli spaces and group actions),
- and the thesis *Equivalence of group actions on Riemann surfaces* [Carvacho Bustamante, 2010].

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Chapter 2

Systoles on the Bolza surface

Within the compact Riemann surfaces of genus two, the so-called *Bolza surface* is a special one. It has the largest automorphism group. This group is $G = \mathrm{GL}_2(3)$ (the group of 2×2 matrices with coefficients in $\mathbb{F}_3$), has order 48 and acts on the Bolza surface with signature $(0; 8, 3, 2)$.

This chapter describes an algorithm to explicitly identify geodesics on a hyperbolic model for the surface, which allows us to find bounds for the kissing number of the Bolza surface B . The procedure follows a clear sequence of steps outlined next.

The idea is to construct a hyperbolic polygon P for B in $\mathbb{D}$. Based on the procedure described in [Behn et al., 2013, sections 3, 4, and 5], we design and implement our own computational routines to build P as well as to measure geodesics on B . This polygon consist of $|G|$ identical copies of a polygon for the quotient surface B/G . These copies are arranged so that G acts by permuting them. This reflects the symmetries on the tiling pattern imposed by G . The action of G is by isometries of $\mathbb{D}$, and these isometries interchange the systoles of B . The size of a G -orbit on a given systole provide a lower bound for the kissing number of B .

The chapter is divided into four main steps. In the first one, in section 2.1, we explain how to construct a hyperbolic polygon Q for the quotient surface B/G , which is isomorphic to the Riemann sphere $\overline{\mathbb{C}}$.

The second one (section 2.2) develops the tiling process of a region P in $\mathbb{D}$, which is guided by choosing a suitable presentation of G . This region is a fundamental polygon for the Bolza surface. To guarantee the existence of an action of $\mathrm{GL}_2(3)$ on B , we show an explicit surface-kernel epimorphism (see [Broughton, 1991, Proposition 2.1]).

The third one, section 2.3, focuses on computing the side pairing of the

curve in the boundary of P . In addition, it addresses a homology basis for the first homology group of B . This basis must be formed from boundary curves, and their geodesic representatives will be simple closed geodesics. By examining the lengths of these geodesic representatives, we obtain upper bounds for the systolic length of B . Should these geodesics, in fact, be systoles? Due to [Schmutz, 1993, Thm 5.2], the answer in this case is yes. In general, they provide us bounds for the systolic length.

In the last step, in section 2.4, we compute the size of the G -orbits of these geodesic representatives. Each such orbit consist of 12 curves, so we obtain a lower bound on the kissing number: there are at least 12 systoles on B . By [Schmutz, 1993, Thm 5.2], these curves are known to be all the systoles of the Bolza surface.

The pictures in the chapter are generated using the Matplotlib Python library in Sage [Stein et al., 2020]. Group-theoretic computations are performed with Sage's interface to GAP [GAP-Group, 2019]. In chapter 4 we collect the full script.

2.1 Step 1: Fundamental polygon for the quotient sphere

A *group action* with signature $(0; 8, 3, 2)$ on $\mathbb{D}$ is a discrete subgroup Δ of isometries of $\mathbb{D}$ such that $\mathbb{D}/\Delta$ is (isomorphic to) the Riemann sphere $\overline{\mathbb{C}}$ and Δ has presentation

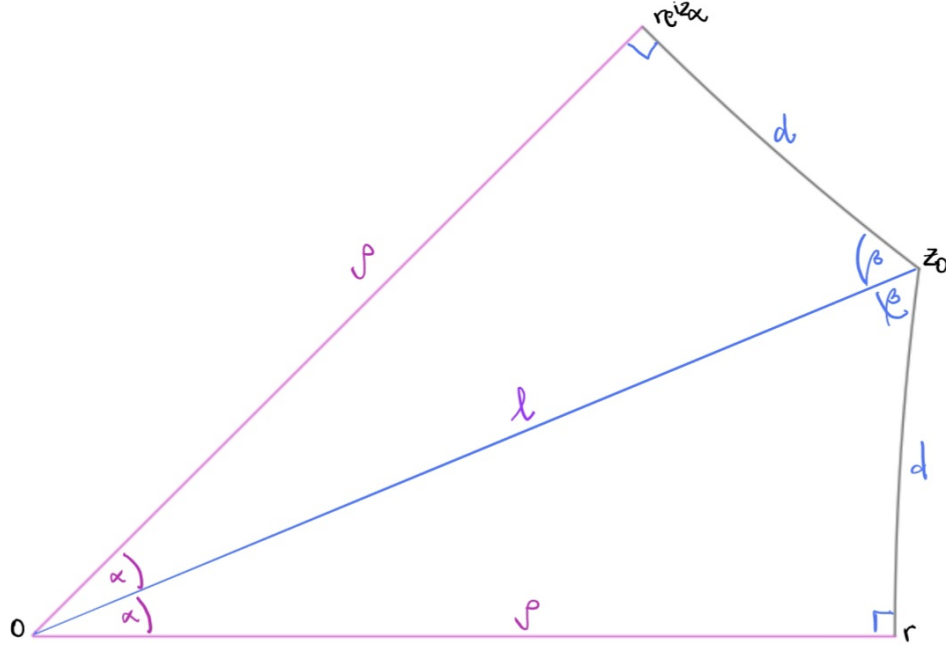
$$\langle X, Y, Z \mid X^8 = Y^3 = Z^2 = XYZ = 1 \rangle = \langle X, Y \mid X^8 = Y^3 = (XY)^2 = 1 \rangle \quad (2.1)$$

Given a concrete group Δ , a *fundamental polygon* for the action of Δ on $\mathbb{D}$ is a polygonal region $D \subseteq \mathbb{D}$. Each Δ -orbit is represented by a unique point in the interior of D , or by finite points in ∂D which are identified by Δ .

One way to choose such a concrete Δ is by taking X as a rotation of order 8 centered at the origin and Y as a rotation of order 3 centered at a point $z_0 \in \mathbb{D}$. By Gauss-Bonnet theorem [Anderson, 2006, Thm 5.16], that fundamental polygon for Δ could not be a triangle with angles $\frac{2\pi}{8}$, $\frac{2\pi}{3}$ and $\frac{2\pi}{2}$, because its area should be $\pi - \left(\frac{2\pi}{8} + \frac{2\pi}{3} + \frac{2\pi}{2} \right) < 0$.

Following [Behn et al., 2013, section 3], a feasible polygon is a quadrilateral Q with angles $\frac{2\pi}{8}$, $\frac{\pi}{2}$, $\frac{2\pi}{3}$, $\frac{\pi}{2}$. We choose a quadrilateral with vertices

$0, r, z_0, re^{i2\alpha}$ (r a real number in $(0, 1)$), the angle at the origin is $2\alpha = \frac{2\pi}{8}$, the angle at z_0 is $2\beta = \frac{2\pi}{3}$, and its edges are hyperbolic geodesics of lengths ρ, d, d, ρ . The number ℓ is the hyperbolic distance from z_0 to the origin. The quadrilateral Q , with the given labels, is shown in figure 2.1.

Figure 2.1: Quadrilateral Q

Using the *hyperbolic trigonometric laws* [Anderson, 2006, Section 5.6], we can compute the number r and the point z_0 . By the hyperbolic cosine law,

$$\begin{aligned}\cos(\beta) &= -\cos(\alpha)\cos(\pi/2) + \sin(\alpha)\sin(\pi/2)\cosh(\rho) \\ \cos(\alpha) &= -\cos(\beta)\cos(\pi/2) + \sin(\beta)\sin(\pi/2)\cosh(d),\end{aligned}$$

and then,

$$\cosh(\rho) = \frac{\cos(\beta)}{\sin(\alpha)}, \quad (2.2)$$

$$\cosh(d) = \frac{\cos(\alpha)}{\sin(\beta)}. \quad (2.3)$$

Therefore, the numbers ρ and d are explicitly

$$\rho = \cosh^{-1} \left(\frac{\cos(\pi/3)}{\sin(\pi/8)} \right), \quad (2.4)$$

$$d = \cosh^{-1} \left(\frac{\cos(\pi/8)}{\sin(\pi/3)} \right) \quad (2.5)$$

The formula for the hyperbolic distance to the origin [Anderson, 2006, Section 4.1] allows us to compute r in terms of the angles α and β .

$$\begin{aligned} \rho &= \log \left(\frac{1+r}{1-r} \right), \\ \cosh(\rho) &= \frac{1}{2} \left(\frac{1+r}{1-r} + \frac{1-r}{1+r} \right) \\ \cosh(\rho) &= \frac{1+r^2}{1-r^2} \\ r^2 &= \frac{\cosh(\rho) - 1}{\cosh(\rho) + 1} \\ r^2 &= \frac{\cos(\beta) - \sin(\alpha)}{\cos(\beta) + \sin(\alpha)} \\ r &= \sqrt{\frac{\cos(\beta) - \sin(\alpha)}{\cos(\beta) + \sin(\alpha)}}. \end{aligned} \quad (2.6)$$

Similarly, the real number $R := |z_0|$ can be computed in terms of α and β . The hyperbolic length ℓ satisfies the following relation:

$$\begin{aligned} \cosh(\ell) &= \cosh(\rho) \cosh(d) - \sinh(\rho) \sinh(d) \cos(\pi/2) \\ \cosh(\ell) &= \frac{\cos(\alpha) \cos(\beta)}{\sin(\alpha) \sin(\beta)}, \end{aligned} \quad (2.7)$$

and then, as in equation 2.6

$$\begin{aligned} \cosh(\ell) &= \log \left(\frac{1+R}{1-R^2} \right) \\ R^2 &= \frac{\cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta)}{\cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta)} \\ R &= \sqrt{\frac{\cos(\alpha + \beta)}{\cos(\alpha - \beta)}} \end{aligned} \quad (2.8)$$

Remark 2.1 (Vertices of Q). As $z_0 = Re^{i\alpha}$, the formulas 2.6 for r and 2.8 for R allows us to express the vertices of Q in terms of $\alpha = \frac{\pi}{8}$ and $\beta = \frac{\pi}{3}$, that is, in terms of the signature of the action.

With these explicit formulas for the vertices, we implement the function `draw_Polygon_mn2` to draw the quadrilateral Q , see section 4.3.

The quadrilateral Q is a fundamental polygon for the action of the group Δ generated by two Möbius transformations. These transformations are represented by matrices X and Y as fractional linear transformations.

$$X = \begin{pmatrix} e^{i2\alpha} & 0 \\ 0 & 1 \end{pmatrix} \quad (2.9)$$

$$Y = \begin{pmatrix} 1 & z_0 \\ \bar{z}_0 & 1 \end{pmatrix} \begin{pmatrix} e^{i2\beta} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -z_0 \\ -\bar{z}_0 & 1 \end{pmatrix} = \begin{pmatrix} e^{i2\beta} - R^2 & z_0(1 - e^{i2\beta}) \\ -\bar{z}_0(1 - e^{i2\beta}) & 1 - e^{i2\beta}R^2 \end{pmatrix}. \quad (2.10)$$

Geometrically, the transformation X is a rotation of order 8 centered at the origin. The transformation Y is a rotation of order 3 centered at z_0 . The three matrices multiplied in Y acts in the following sequence: the Möbius transformation represented by $\begin{pmatrix} 1 & -z_0 \\ -\bar{z}_0 & 1 \end{pmatrix}$ sends z_0 to 0. Then, $\begin{pmatrix} e^{i2\beta} & 0 \\ 0 & 1 \end{pmatrix}$ performs an order 3 rotation around 0. The last matrix returns 0 to z_0 .

Remark 2.2 (Side pairing of Q). Denote by $\gamma(p, q)$ the hyperbolic geodesic starting at p and ending at q . The transformation X sends the geodesic $\gamma(0, r)$ to $\gamma(0, re^{i2\alpha})$, while Y sends $\gamma(r, z_0)$ to $\gamma(re^{i2\alpha}, z_0)$, so the quotient sphere $\bar{\mathbb{C}}$ results from the identification of $\gamma(0, r)$ with $\gamma(0, re^{i2\alpha})$ and $\gamma(r, z_0)$ with $\gamma(re^{i2\alpha}, z_0)$.

In the presentation (2.1), the element XY must have order 2. Applying the following lemma, we can ensure that this condition is satisfied.

Lemma 2.3. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be an invertible matrix with complex coefficients. If $a + d = 0$, the Möbius transformation represented by M has order 2.

Proof. Let M be a matrix of the form $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$, where a, b, c are complex numbers such that $a^2 + bc \neq 0$. Then,

$$M^2 = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \begin{pmatrix} a & b \\ c & -a \end{pmatrix} = \begin{pmatrix} a^2 + bc & 0 \\ 0 & a^2 + bc \end{pmatrix}. \quad (2.11)$$

The Möbius transformation represented by M^2 is the identity function. $\square$

Our matrix XY is

$$XY = \begin{pmatrix} e^{i2\alpha}(e^{i2\beta} - R^2) & e^{i2\alpha}z_0(1 - e^{i2\beta}) \\ -\bar{z}_0(1 - e^{i2\beta}) & 1 - e^{i2\beta}R^2 \end{pmatrix} \quad (2.12)$$

and the sum of coefficients required by lemma 2.3 is

$$(XY)_{11} + (XY)_{22} = e^{i2\alpha}(e^{i2\beta} - R^2) + 1 - e^{i2\beta}R^2.$$

Computing the real and imaginary parts separately,

$$\begin{aligned} \operatorname{Re}((XY)_{11} + (XY)_{22}) &= \cos(2\alpha + 2\beta) - \cos(2\alpha)R^2 + 1 - \cos(2\beta)R^2 \\ &= 1 + \cos(2\alpha + 2\beta) - (\cos(2\alpha) + \cos(2\beta)) \frac{\cos(\alpha + \beta)}{\cos(\beta - \alpha)} \\ &= 1 + 2\cos(\alpha + \beta)^2 - 1 \\ &\quad - 2\cos(\alpha + \beta)\cos(\beta - \alpha) \frac{\cos(\alpha + \beta)}{\cos(\beta - \alpha)} \\ &= 2\cos(\alpha + \beta)^2 - 2\cos(\alpha + \beta)^2 \\ \operatorname{Re}((XY)_{11} + (XY)_{22}) &= 0, \end{aligned} \quad (2.13)$$

$$\begin{aligned} \operatorname{Im}((XY)_{11} + (XY)_{22}) &= \sin(2\alpha + 2\beta) - \sin(2\alpha)R^2 - \sin(2\beta)R^2 \\ &= \sin(2\alpha + 2\beta) - (\sin(2\alpha) + \sin(2\beta)) \frac{\cos(\alpha + \beta)}{\cos(\beta - \alpha)} \\ &= \sin(2\alpha + 2\beta) - 2\cos(\beta - \alpha)\sin(\alpha + \beta) \frac{\cos(\alpha + \beta)}{\cos(\beta - \alpha)} \\ &= \sin(2\alpha + 2\beta) - 2\sin(\alpha + \beta)\cos(\alpha + \beta) \\ \operatorname{Im}((XY)_{11} + (XY)_{22}) &= 0. \end{aligned} \quad (2.14)$$

As $(XY)_{11} + (XY)_{22} = 0$, the Möbius transformation XY has order 2.

In the next section, we develop the second step: the construction of a fundamental polygon for the Bolza surface: a genus two surface endowed with an action of $\operatorname{GL}_2(3)$ with signature $(0; 8, 3, 2)$.

2.2 Step 2: Fundamental polygon for the surface

By the Riemann-Hurwitz formula, the group $\operatorname{GL}_2(3)$ could act with signature $(0; 8, 3, 2)$ on a genus two surface. Let Γ be an abstract Fuchsian group uniformizing such a genus two surface. This group Γ must have the following presentation.

$$\Gamma \cong \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1] \cdot [a_2, b_2] = 1 \rangle$$

In addition to the Riemann-Hurwitz formula, Riemann's existence theorem [Broughton, 1991, Proposition 2.1] requires a *generating vector* (a, b, c) of $\mathrm{GL}_2(3)$. This generating vector comes from a group epimorphism θ .

$$\begin{aligned}\theta: \Delta &\rightarrow \mathrm{GL}_2(3) \\ X &\rightarrow a \\ Y &\rightarrow b \\ XY &\rightarrow c.\end{aligned}$$

The group Δ has presentation (2.1). By θ , the element a must have order 8, the element b order 3, and c order 2. In addition, these elements must be generators of $G = \mathrm{GL}_2(3)$. Next, using GAP [GAP-Group, 2019], we obtain all possible triples of elements (a, b, c) of G as follows.

First, we collect all elements of G in three sublists: the first sublist contains elements of order 8, the second contains elements of order 3, and the last, elements of order 2 (see `search_generators` in section 4.2).

Next, we iterate over the output list of `search_generators`. We select one element from each sublist. If the chosen triple generates the group G , we add it to a list `generators` of admissible generators of G (see `potencial_generators` in section 4.2).

Since in our context all the $\mathrm{GL}_2(3)$ -actions with signature $(0; 8, 3, 2)$, we choose the first triple (a, b, c) obtained from `potencial_generators` as a generating vector for G . We refer the reader to [Carvacho Bustamante, 2010] for more on equivalence of group actions, particularly to chapter two and the reference within it.

Finally, through the GAP interface in Sage, we apply an automorphism of $\mathrm{GL}_2(3)$ to change its presentation from the given in the GAP catalog to one whose generators are precisely a, b , and c (see section 4.2). We denote by K the finitely presented group isomorphic to $\mathrm{GL}_2(3)$ with the desired presentation, which is explicitly

$$K = \langle x, y, z \mid x^8 = z^2 = yzx = y^3 = (x^{-1}y^{-1}zx^{-1})^2 = zx^{-3}y^{-1}x^4y^{-1} = 1 \rangle, \quad (2.15)$$

and in terms of the generators x, y, z its elements are

$$K = \{x^j, x^jy, x^jxyx, x^jxyx^2, x^jxyx^3, x^jxyx^3y \mid 0 \leq j \leq 7\}. \quad (2.16)$$

With the concrete Fuchsian group Δ generated by X as in (2.9) and Y

as in (2.10), we define the concrete group epimorphism

$$\begin{aligned} \theta: \Delta &\rightarrow K \\ X &\mapsto x \\ Y &\mapsto y \\ Z &\mapsto z. \end{aligned} \tag{2.17}$$

Let Γ be the kernel of θ . This group is the normal closure on Δ of the subgroup generated by the set of relations defining K that are not in the presentation (2.1) of Δ . These extra relations make the quotient group Δ/Γ into a finite group isomorphic to G . The generating relations of Γ are the following.

$$\Gamma = \langle (X^{-1}Y^{-1}ZX^{-1})^2, ZX^{-3}Y^{-1}X^4Y^{-1} \rangle \tag{2.18}$$

By covering map theory [Greenberg and Harper, 2018, section 5.8], the group Γ is isomorphic to the fundamental group of the Bolza surface B , hence B is isomorphic to $\mathbb{D}/\Gamma$. A fundamental polygon for $\mathbb{D}/\Gamma$ in $\mathbb{D}$ consist of 48 copies of the fundamental polygon of $\mathbb{D}/\Delta$, one copy of each Γ -coset in Δ (see [Behn et al., 2013, section 4] and the references given there).

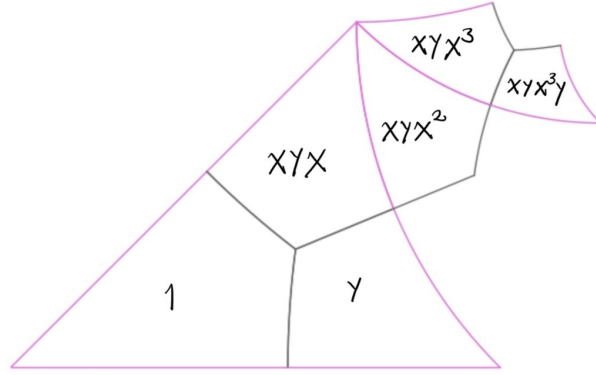
Let K_Δ be the following set of Γ -cosets representatives in Δ (compare with (2.16)). We compute and store the matrices in (2.19) with the function `as_matrices_mn2` (see section 4.2).

$$K_\Delta = \{X^j, X^jY, X^jXYX, X^jXYX^2, X^jXYX^3, X^jXYX^3Y \mid 0 \leq j \leq 7\}. \tag{2.19}$$

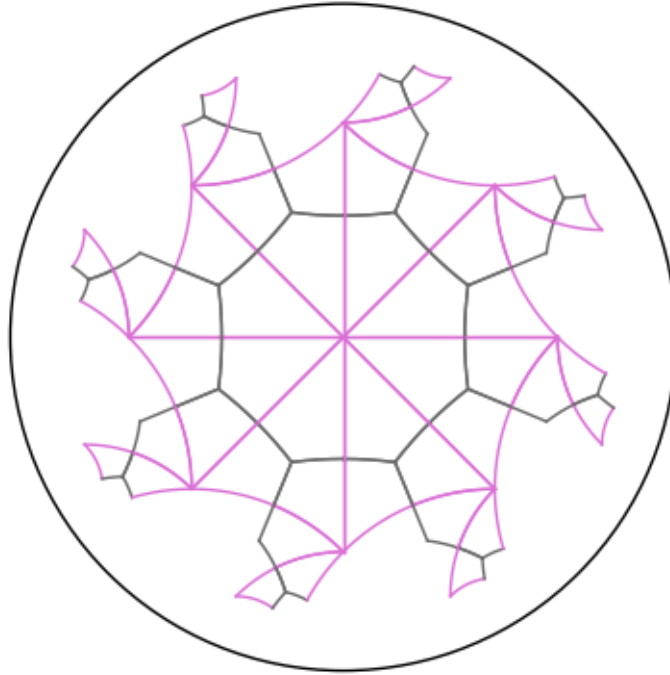
In section 2.1, we found a fundamental quadrilateral Q for $\mathbb{D}/\Delta$. To build a fundamental polygon P of B , take images of Q under all Möbius transformations in (2.19) to define the polygonal region

$$P = \bigcup_{W \in K_\Delta} W(Q), \tag{2.20}$$

Taking $j = 0$ in (2.19), we obtain the polygon depicted in figure 2.2. Each image of Q is labeled with the corresponding Möbius transformation applied to it. The pink sides have length ρ (with ρ as in (2.4)) and the gray ones have length d (with d as in (2.5)).

Figure 2.2: Eighth part of P

The complete polygon P is drawn using `fundamentalpolygon_mn2` (see section 4.3), inside the unit disc $\mathbb{D}$ in figure 2.2.

Figure 2.3: Polygon P inside $\mathbb{D}$

By the Gauss-Bonnet theorem, the area of the quadrilateral Q is

$$A(Q) = (4 - 2)\pi - \left(\frac{2\pi}{8} + \frac{2\pi}{3} + \frac{\pi}{2} + \frac{\pi}{2} \right) = 2\pi \left(1 - \left(\frac{1}{8} + \frac{1}{3} + \frac{1}{2} \right) \right),$$

and by the Riemann-Hurwitz formula, the right-hand side can be written as

$$\begin{aligned} 2g(B) - 2 &= |G| \left(-2 + 3 - \left(\frac{1}{8} + \frac{1}{3} + \frac{1}{2} \right) \right) \\ \frac{2}{|G|} &= 1 - \left(\frac{1}{8} + \frac{1}{3} + \frac{1}{3} \right). \end{aligned}$$

Thus, the area of Q is $A(Q) = 2\pi \cdot 2/|G|$, and the polygon P has area

$$A(P) = |G| \cdot A(Q) = 4\pi. \quad (2.21)$$

Recall our objective: to find bounds for the kissing number of B . One key step, of course, is finding candidates for systoles on the surface.

According to Schmutz-Schaller [Schmutz, 1993, Thm 5.2], among genus two surfaces of area 4π , the Bolza surface has the longest systoles with length $2 \cosh^{-1}(1 + \sqrt{2})$. Some algebraic manipulations show that 2ρ , with ρ as in (2.4), is half this value:

$$\begin{aligned} \cosh(\rho) &= \frac{\sqrt{2 + \sqrt{2}}}{\sqrt{2}}, \\ \cosh(\rho)^2 &= \frac{2 + \sqrt{2}}{2} \\ \frac{e^{2\rho} + e^{-2\rho} + 2}{2} &= 2 + \sqrt{2} \\ \cosh(2\rho) &= 1 + \sqrt{2} \\ 2\rho &= \cosh^{-1}(1 + \sqrt{2}) \end{aligned}$$

Remark 2.4 (Systoles of the Bolza surface). *The closed geodesics on P of length $4\rho = 2 \cosh^{-1}(1 + \sqrt{2}) \approx 3.0571$ are the candidates to be systoles.*

Since P is simply connected, we expect to find systoles among closed geodesics with endpoints at the boundary of P . These geodesics represent a homology class of boundary curves.

In the next section, which is the third step of our strategy, we compute a homology basis for B . A posteriori, the geodesic representatives of the homology class of the generators will be systoles.

2.3 Step 3: Side pairing and homology basis

Each boundary curve α of P is contained in two different images of Q under elements of Δ . One image is inside the polygon P . The other lies outside P .

Denote by I_α the element of Δ in K_Δ (see 2.19), that is, the element such that $I_\gamma(Q) \subseteq P$, and by O_α the element of Δ not in K_Δ such that $O_\alpha \cap P = \alpha$. The notation comes from the words *inner* and *outer* to refer to the position of images of Q with respect to P .

Given a boundary curve γ , the polygon $O_\gamma(Q)$ lies in the same Γ -coset of a polygon $M(Q)$, with M in K_Δ . If $L \in \Gamma$ is the element such that $LO_\gamma = M$, we observe that the curve $L(\gamma)$ is a boundary curve of P .

The side pairing of P is an equivalence relation on boundary curves which captures the difference by an element of Γ . More precisely, two boundary curves α and β are *side-paired* if there exist an element $L \in \Gamma$ such that $I_\alpha = LO_\beta$.

For example, the geodesic γ starting at $Y(0)$ and ending at $Y(r)$ (r as in (2.6)) is contained in the polygon $YX^{-1}(Q)$, so $I_\gamma = Y$ and $O_\gamma = YX^{-1}$ (see figure 2.3).

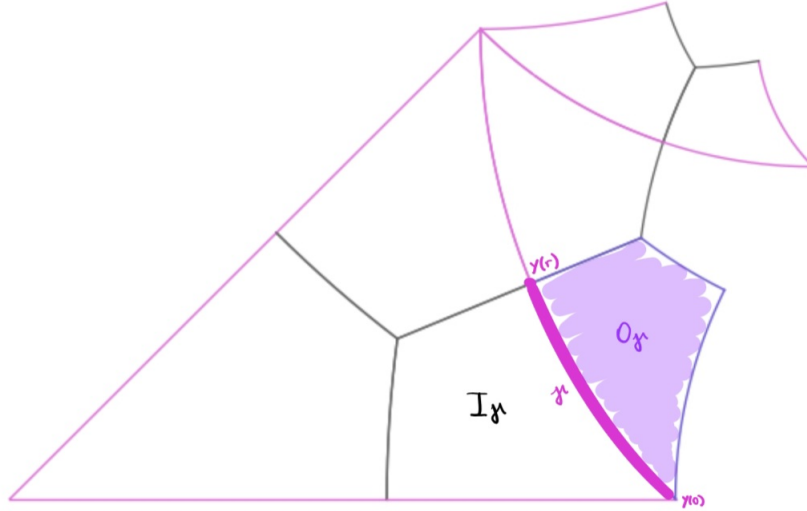


Figure 2.4: Eighth part of P with the extra polygon $O_\gamma(Q)$

The side pairing relation states that the curve γ' paired to γ satisfies $I_{\gamma'} = MO_\gamma$ for some $M \in \Gamma$. Using GAP groups' method **Factorization**,

we express $I_{\gamma'}$ as X^4YX^3 . The curve γ' in figure 2.5 is side-paired to γ .

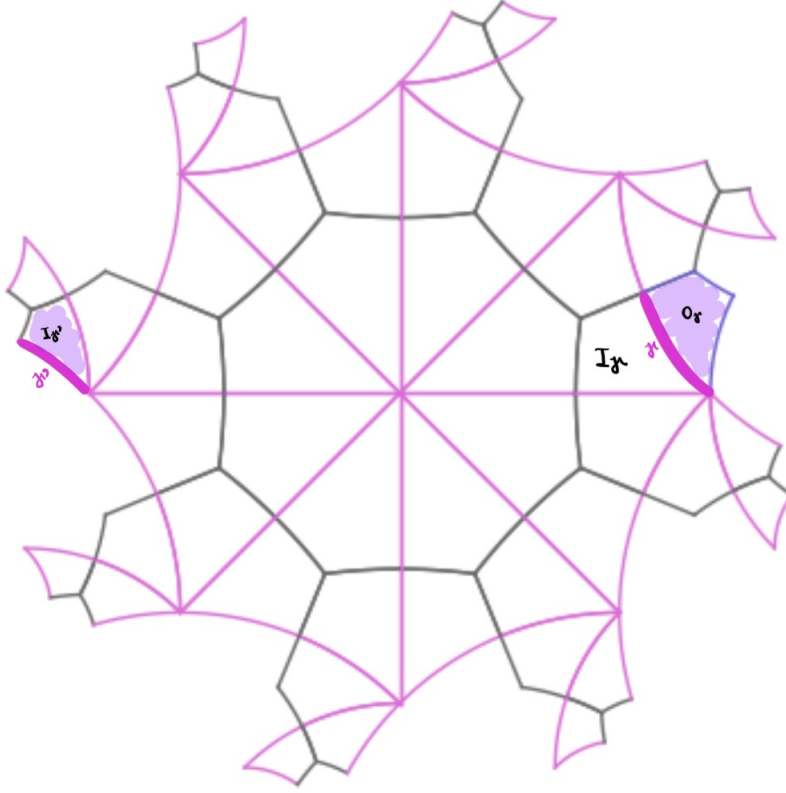
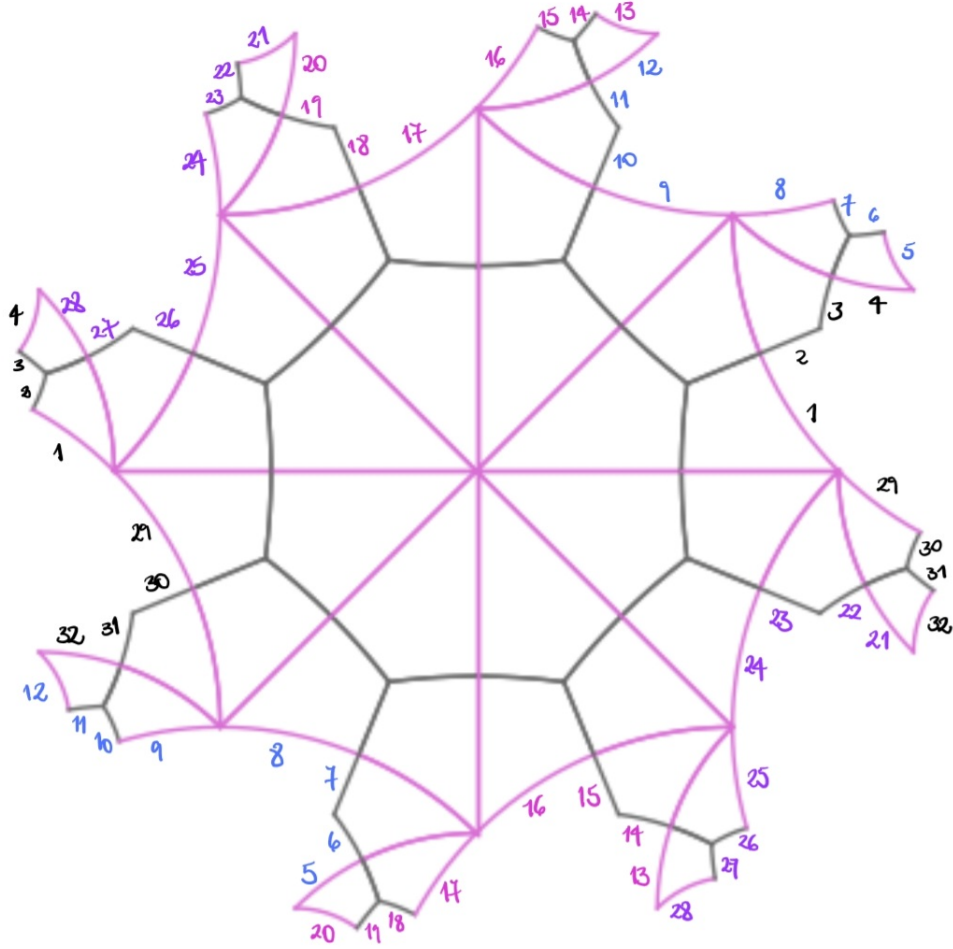
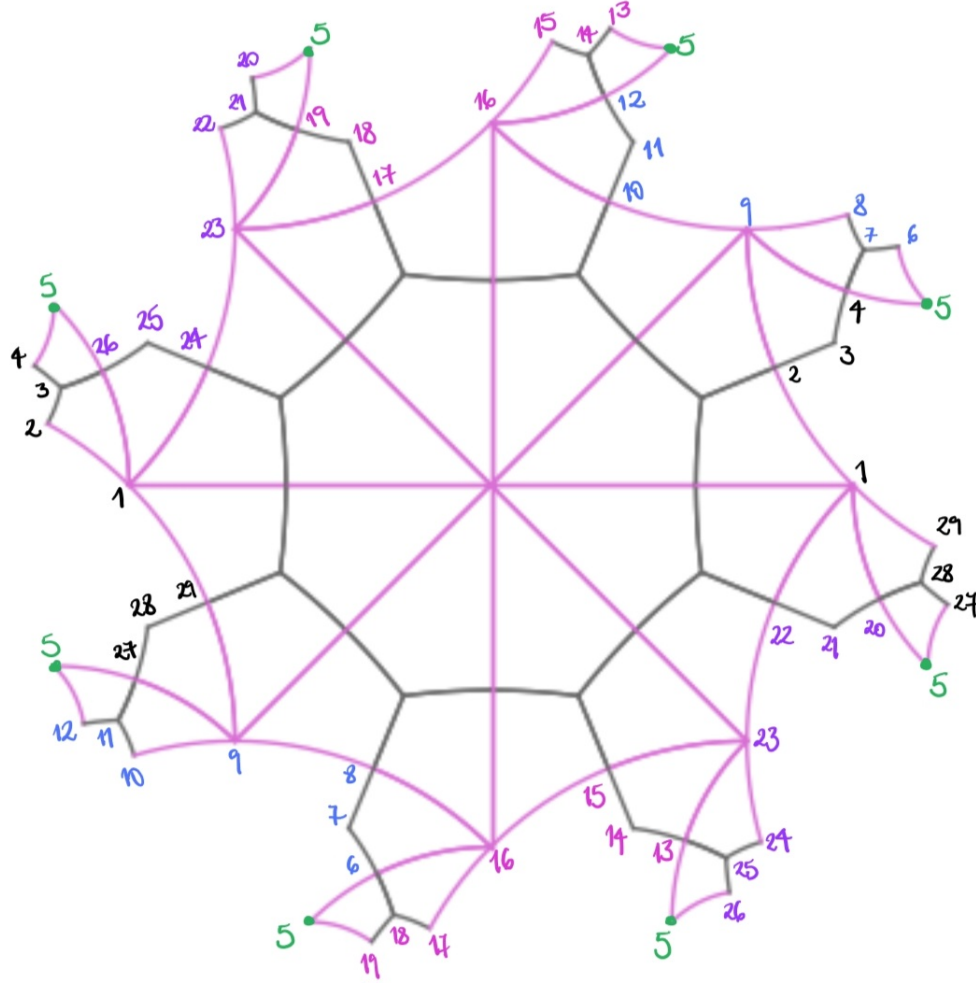


Figure 2.5: Curves γ and γ' are identified.

Applying the same procedure along all boundary curves, we complete the side pairing of P . The curves in the same class receive the same label (see figure 2.6).

Figure 2.6: Side pairing of B

As a result of the side pairing, the boundary of P has 32 edges and 29 vertices. The vertices are depicted in figure 2.7.

Figure 2.7: Boundary vertices of B

To determine a homology basis for B , we use the following exact sequence in relative homology with respect to the set of boundary vertices V of P ([Behn et al., 2013, lemma 5.1] and [Greenberg and Harper, 2018, section 14]).

$$0 \longrightarrow H_1(B) \xrightarrow{j} H_1(B, V) \xrightarrow{\bar{\partial}} H_0(V) \xrightarrow{i} H_0(B) \longrightarrow 0 \quad (2.22)$$

Let $V = \{P_1, P_2, \dots, P_{32}\}$ with P_k the vertex labeled as k in figure 2.7, and let $E = \{\gamma_1, \gamma_2, \dots, \gamma_{32}\}$ be the set of boundary edges. Given an abstract

closed curve α , which represents a class in $H_1(B)$, the injectivity of the map j allows us to identify α with a curve with endpoints in A . The curve $j(\alpha)$ can be expressed as an integral combination of the form:

$$j(\alpha) = \sum_{j=1}^{32} a_j \gamma_j.$$

The connecting homomorphism $\bar{\partial}$ in the sequence (2.22) is exactly the boundary operator on $j(\alpha)$. Then,

$$\begin{aligned} \bar{\partial}j(\alpha) &= \sum_{k=1}^{32} a_k \partial(\gamma_k) \\ &= \left(\sum_{k=1}^{11} a_k (P_{k+1} - P_k) \right) + a_{12}(P_5 - P_{12}) + a_{13}(P_{13} - P_5) + \left(\sum_{k=14}^{19} a_k (P_k - P_{k-1}) \right) \\ &\quad + a_{20}(P_5 - P_{19}) + a_{21}(P_{20} - P_5) + \left(\sum_{k=22}^{27} a_k (P_{k-1} - P_{k-2}) \right) \\ &\quad + a_{28}(P_5 - P_{26}) + a_{29}(P_1 - P_{27}) + a_{30}(P_{27} - P_{28}) + a_{31}(P_{28} - P_{29}) + a_{32}(P_{29} - P_5). \end{aligned}$$

Collecting coefficients, and by the exactness of the sequence (2.22), we obtain the following linear system. The colors of the equations indicates from which set of vertices they come, see the color labels in figure 2.7.

$$-a_{31} + a_{32} = -a_{30} + a_{31} = -a_{29} + a_{30} = -a_1 + a_{29} = a_1 - a_2 = a_2 - a_3 = a_3 - a_4 = 0 \quad (2.23)$$

$$a_4 - a_5 + a_{12} - a_{13} + a_{20} - a_{21} + a_{28} - a_{32} = 0 \quad (2.24)$$

$$a_5 - a_6 = a_6 - a_7 = a_7 - a_8 = a_8 - a_9 = a_9 - a_{10} = a_{10} - a_{11} = a_{11} - a_{12} = 0 \quad (2.25)$$

$$a_{13} - a_{14} = a_{14} - a_{15} = a_{15} - a_{16} = a_{16} - a_{17} = a_{17} - a_{18} = a_{18} - a_{19} = a_{19} - a_{20} = 0 \quad (2.26)$$

$$a_{21} - a_{22} = a_{22} - a_{23} = a_{23} - a_{24} = a_{24} - a_{25} = a_{25} - a_{26} = a_{26} - a_{27} = a_{27} - a_{28} = 0. \quad (2.27)$$

The equation (2.24) is redundant, and the others are equivalent to the following equalities.

$$a_1 = a_2 = a_3 = a_4 = a_{29} = a_{30} = a_{32} = a_{32} \quad (2.28)$$

$$a_5 = a_6 = a_7 = a_8 = a_9 = a_{10} = a_{11} = a_{12} \quad (2.29)$$

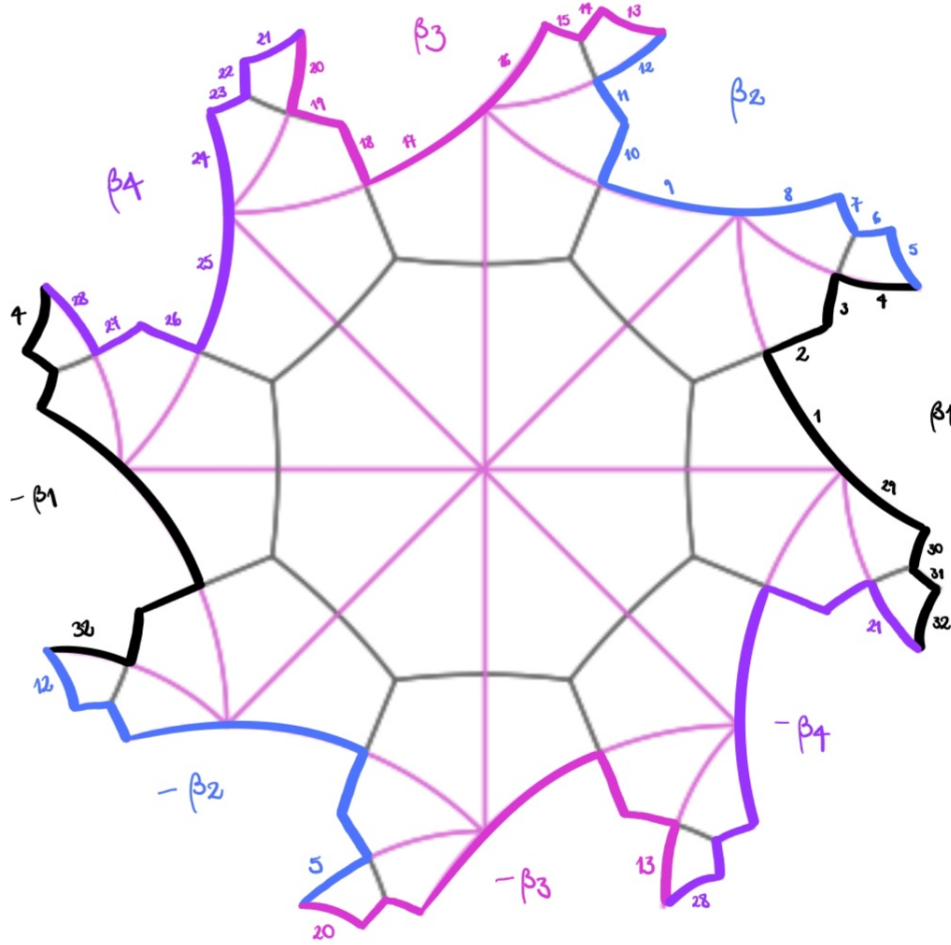
$$a_{13} = a_{14} = a_{15} = a_{16} = a_{17} = a_{18} = a_{19} = a_{20} \quad (2.30)$$

$$a_{21} = a_{22} = a_{23} = a_{24} = a_{25} = a_{26} = a_{27} = a_{28}. \quad (2.31)$$

Therefore, the relative homology class of α is determined by four integral parameters: the values of a_1, a_5, a_{13} and a_{21} . Denote by v_α the corresponding integral vector of α .

Take an abstract basis $\mathcal{B} = \{\beta_1, \beta_2, \beta_3, \beta_4\}$ of $H_1(B)$. As j is injective, the vectors $v_{\beta_1}, v_{\beta_2}, v_{\beta_3}$ and v_{β_4} must be linearly independent. We choose the basis with $v_{\beta_1} = (1, 0, 0, 0), v_{\beta_2} = (0, 1, 0, 0), v_{\beta_3} = (0, 0, 1, 0)$ and $v_{\beta_4} = (0, 0, 0, 1)$. The corresponding boundary curves are illustrated in figure 2.8: the curve β_1 in black, β_2 in blue, β_3 in pink, and β_4 in violet.

By the side pairing, these four curves are identified to other boundary curves. In figure 2.8, the side-paired curve is marked with a negative sign to indicate the orientation.

Figure 2.8: Homology basis for B

Remark 2.5. The curves $\beta_1, \beta_2, \beta_3$ and β_4 have the same hyperbolic length $4\rho + 4d$, with ρ as in (2.4) and d as in (2.5).

However, the geodesic representatives in the homotopy class of these curves have length 4ρ .

Let α_j be the geodesic representative of the homotopy class of β_j for $j \in \{1, 2, 3, 4\}$. In figure 2.9, the curve α_1 is the black one, α_2 the blue one, α_3 the pink one, and α_4 the violet.

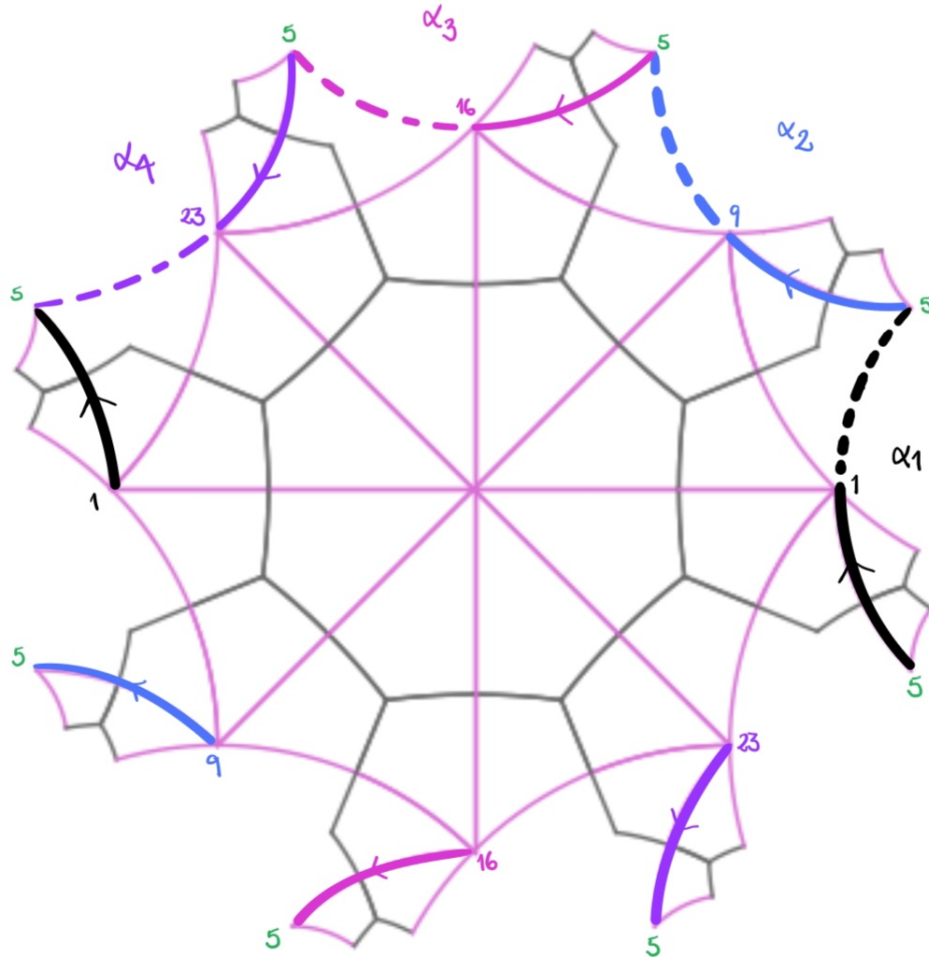


Figure 2.9: Geodesic representatives of the found homology basis

To illustrate this further, consider the black geodesic loop on the right. It is formed by two geodesic segments of lengths 2ρ . The first half (continuous black) lies inside the polygon P , while the second half (dashed) is outside P . In the side pairing, the dashed black geodesic is identified with the left continuous black curve.

Although α_1 seems to be discontinuous, both halves are glued by the vertex 1. The same phenomenon is observed on the curves α_2, α_3 , and α_4 .

These four curves are closed geodesics with length 4ρ , so they provide an upper bound for $\text{sys}(B)$. Nevertheless, in this case, by remark 2.4 they are actually systoles.

Based on this observation, the next and final section focuses on the kissing number of B . Specifically, starting with the four systoles just found, we will compute its orbits under an action of $\mathrm{GL}_2(3)$.

2.4 Step 4: Group action on geodesic representatives

Each Möbius transformations in K_Δ (see (2.19)) acts on $\mathbb{D}$ by isometries. Then, geodesics of length 4ρ are sent to geodesics of length 4ρ (ρ as in (2.4)). If we start with such a geodesic inside P , its image may lie outside P . However, the image curve is in the same Γ -coset (with Γ as in (2.18)) as another curve inside P . With this observation, we define an action of $\mathrm{GL}_2(3)$ on P as follows.

Given $h \in \mathrm{GL}_2(3)$, denote by $\bar{h}$ the unique preimage of g under θ in K_Δ (see (2.17)). Each point $p_0 \in P$ belongs to a quadrilateral $W_{p_0}(Q)$, for some $W_{p_0} \in K_\Delta$ (see (2.20)). Explicitly, we consider the following (left) action

$$\begin{aligned} \mathrm{GL}_2(3) \times P &\rightarrow P \\ (h, p_0) &\mapsto h \cdot p := \overline{h\theta(W_{p_0})}(p). \end{aligned} \quad (2.32)$$

For example, the generator y of $\mathrm{GL}_2(3)$ sends the quadrilateral $Y(Q)$ to $XYX(Q)$ (with X the Möbius transformation given by (2.9) and Y by (2.10)).

Each element k of $\mathrm{GL}_2(3)$ induces a linear homomorphism $H_1(k)$ on $H_1(B)$; we will denote also as $H_1(k)$ the representing matrix with respect to the basis $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ of $H_1(B)$ (see figure 2.9). As $\mathrm{GL}_2(3)$ is finitely generated, we only require the matrices $H_1(x)$ and $H_1(y)$ to describe the whole action of $\mathrm{GL}_2(3)$ on $H_1(B)$.

The element x acts on P by an order 8 counterclockwise rotation around the origin. In figure 2.9 we observe that $x \cdot \alpha_1 = \alpha_2$, $x \cdot \alpha_2 = \alpha_3$, $x \cdot \alpha_3 = \alpha_4$, and $x \cdot \alpha_4 = \alpha_1$ because of the orientation. Therefore, the matrix representing the x -action is

$$H_1(x) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (2.33)$$

is the matrix representing $H_1(x)$.

For the action of y , we separate the geodesics into four segments of length ρ . For instance, the geodesic α_1 starts in an edge of the polygon $YX^3Y(Q)$,

follows in the intersection between $YX^2(Q)$ and $YX^3(Q)$, continues on the intersection between $X^4YX^2(Q)$ and $X^4YX^3(S)$, and ends in $X^4YX^3Y(Q)$.

Using `as_element_in_G` to perform reductions, we obtain that $y \cdot \alpha_1$ consist of an edge of $X^6YX(Q)$, the intersection between $XYX^3(Q)$ and $X^5Y(Q)$, the intersection between $X^5YX^3(Q)$ and $XY(Q)$, and finishes in $X^2YX(Q)$. By a linear fixed-endpoint homotopy (indicated with green arrows in figure 2.10), the curve $y \cdot \alpha_1$ is homotopic to $\alpha_2 + \alpha_3$.

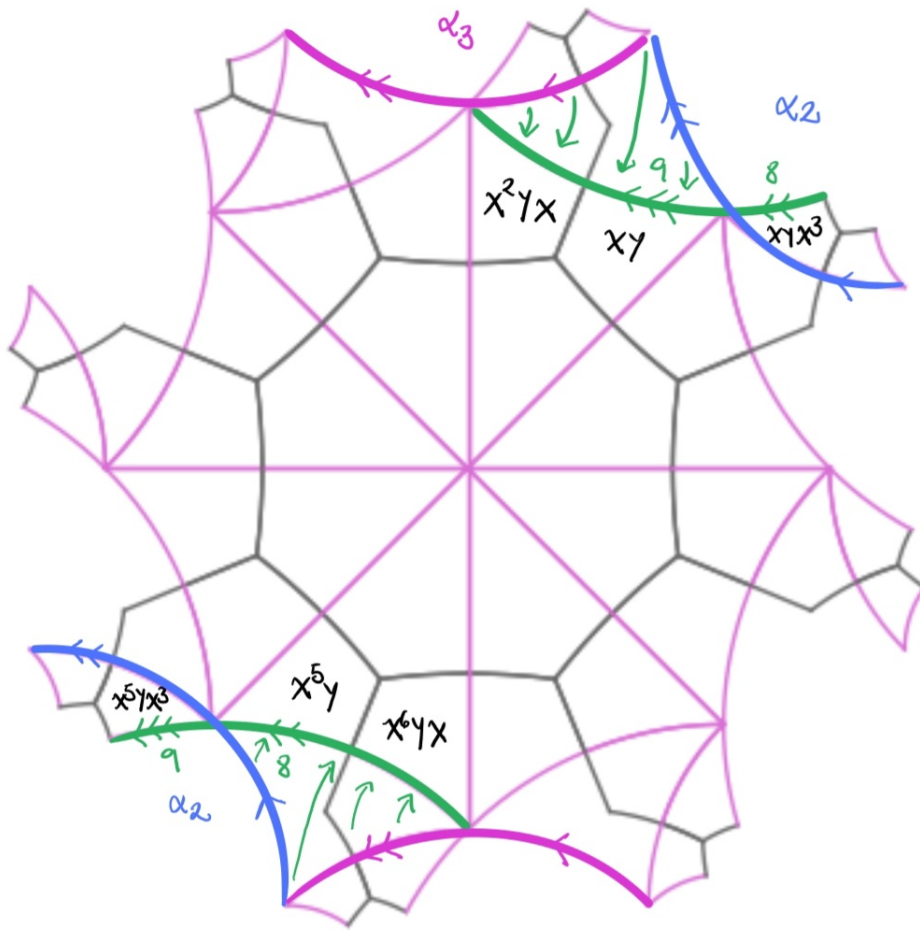


Figure 2.10: Homotopy between $y \cdot \alpha_1$ and $\alpha_2 + \alpha_3$

Analogously, we apply fixed-endpoints if needed to represent the curves $y \cdot \alpha_2$, $y \cdot \alpha_3$, and $y \cdot \alpha_4$ in terms of α_1 , α_2 , α_3 , and α_4 . These curves are,

respectively, depicted in figure 2.11, figure 2.12, and figure 2.13 respectively.

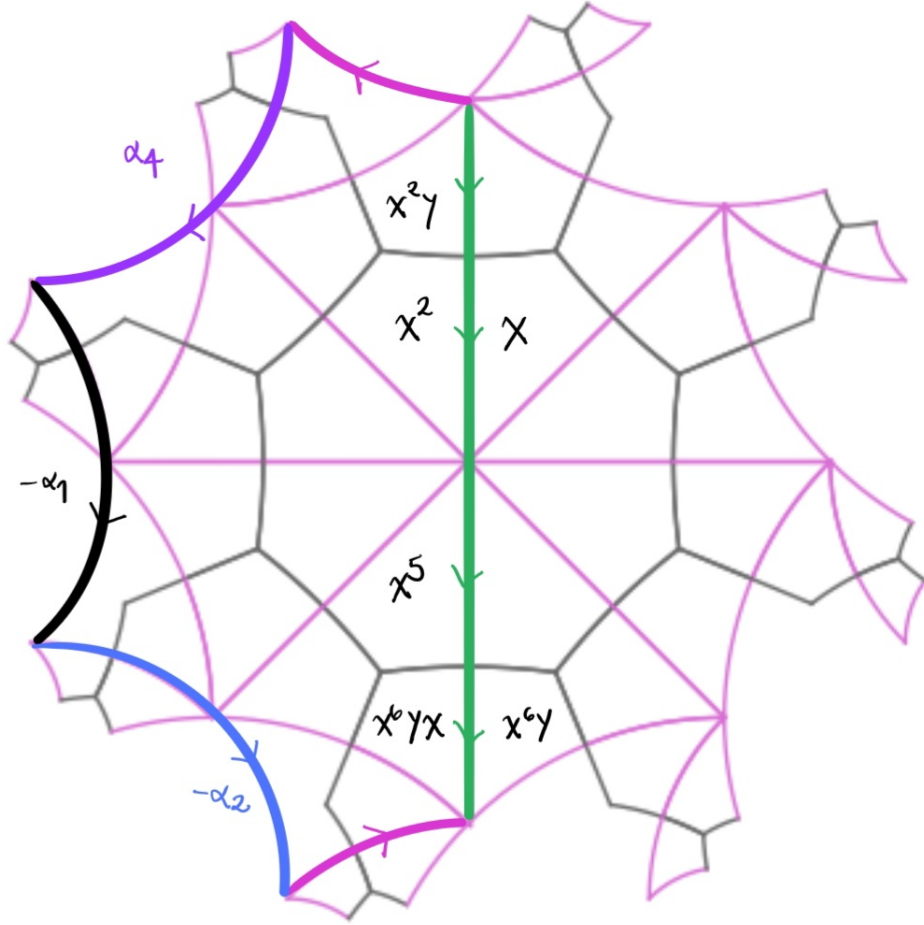


Figure 2.11: Homotopy between $y \cdot \alpha_2$ and $-\alpha_1 - \alpha_2 + \alpha_4$. The pink curves are side-paired, but with different orientations. So, they canceled out.

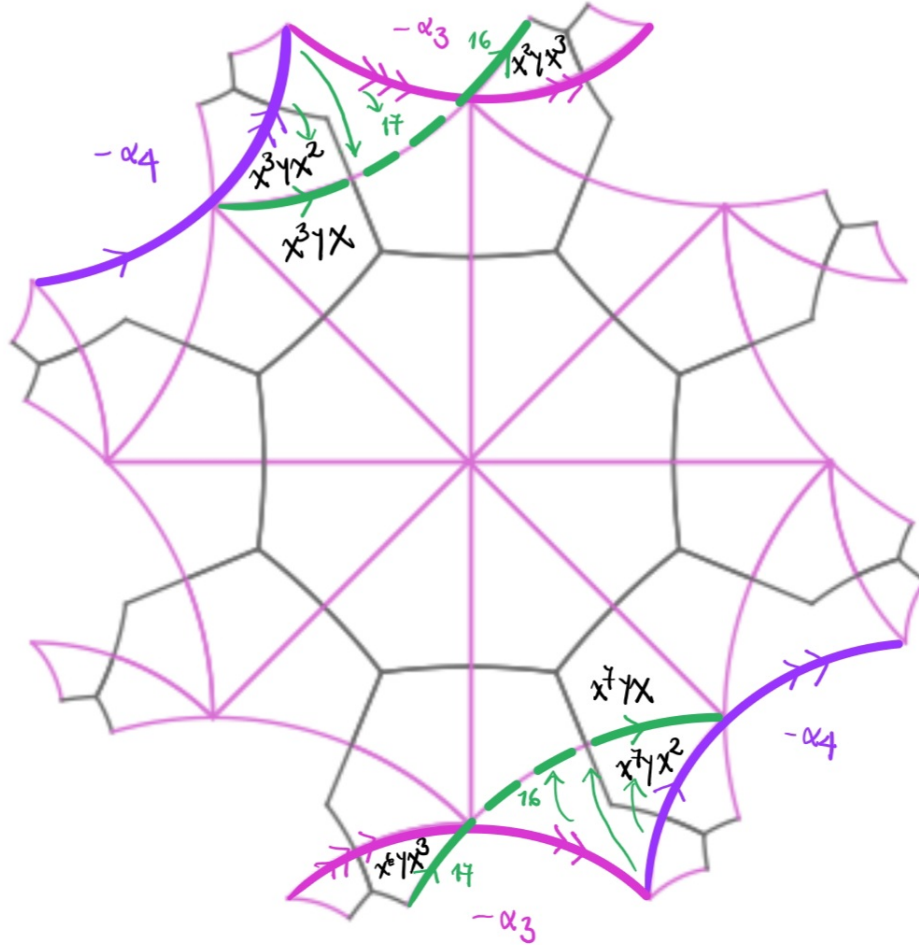


Figure 2.12: Homotopy between $y \cdot \alpha_3$ and $-\alpha_3 - \alpha_4$.

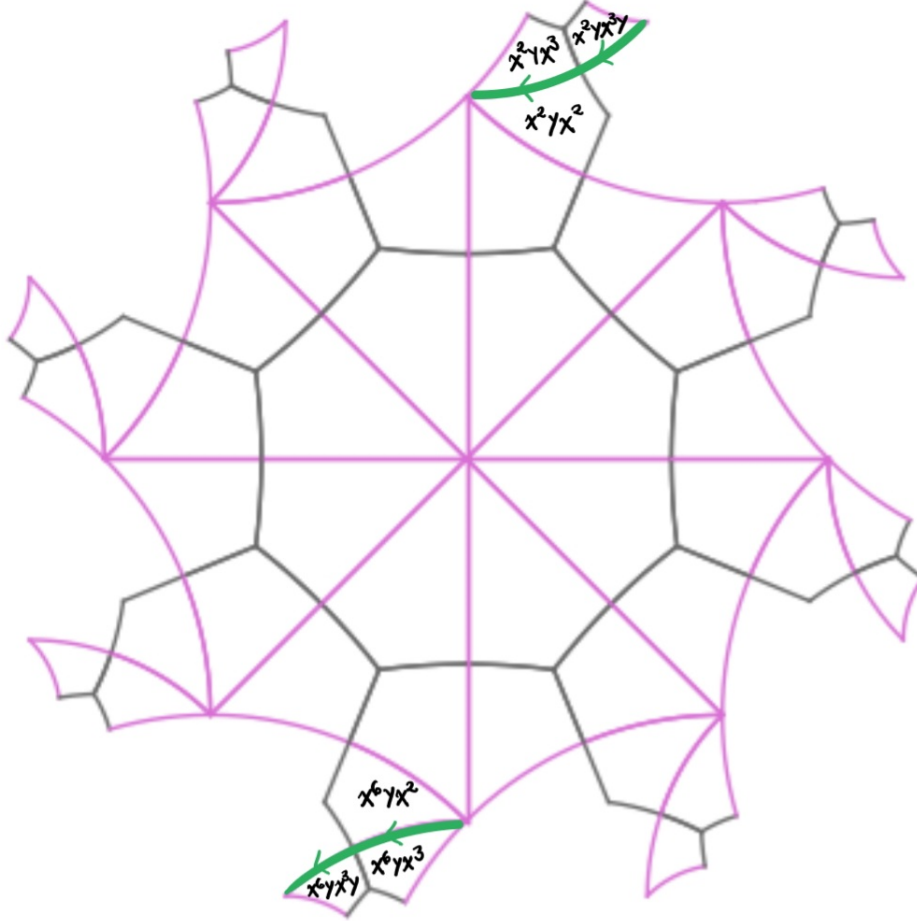


Figure 2.13: The curve $y \cdot \alpha_4$ is exactly α_3 .

The matrix corresponding to the action of y is

$$H_1(y) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix}. \quad (2.34)$$

The action 2.35 on $H_1(B)$ has representation

$$\begin{aligned} \mathrm{GL}_2(3) &\rightarrow \mathrm{GL}(\mathbb{Z}^4) \\ w &\mapsto H_1(w). \end{aligned} \quad (2.35)$$

The image of $\mathrm{GL}_2(3)$ is the subgroup H of $\mathrm{GL}(\mathbb{Z}^4)$ generated by $H_1(x)$ and $H_1(y)$. The explicit matrices in H can be computed using `action_of_G` (see section 4.2).

Finally, we are able to compute the H -orbit of α_1 in $H_1(B)$ using Orbit-Stabilizer theorem. The stabilizer subgroup of the vector $(1, 0, 0, 0)$, corresponding to α_1 , is computed using `stabilizers` (see section 4.2). This group is explicitly $\mathrm{Stab}_H(\alpha_1) = \{1, H_1(x^2yx^3), H_1(x^6yx^3), H_1(x^4)\}$. Therefore,

$$|H \cdot \alpha_1| = \frac{|H|}{|\mathrm{Stab}_H(\alpha_1)|} = 12. \quad (2.36)$$

Theorem 2.6 (Bound on the kissing number). *These 12 curves are geodesics of length 4ρ , so all of them are systoles of the Bolza surface. Then,*

$$\mathrm{Kiss}(B) \geq 12. \quad (2.37)$$

In fact, according to [Schmutz, 1993, Thm 5.2], the Bolza surface has 12 systoles. Therefore, we have found them all explicitly in the hyperbolic polygon uniformizing B .

This method also applies to other surfaces endowed with group action of signature $(0; m, n, 2)$, yielding lower bounds for their kissing number.

In summary, we found explicit curves that realize the bounds of [Schmutz, 1993]. The next table collects our findings.

Quantity	Value	See in
Systolic length	$2 \cosh^{-1}(1 + \sqrt{2})$	Rmk 2.4
Kissing number	12	Thm 2.6

Chapter 3

Families on genus two

On the moduli space of compact Riemann surfaces of genus two, there are surfaces with non-trivial group actions. According to [Broughton, 1991, Table 4] and [Rodriguez and Rojas, 2024, Section 3], within the moduli space there are three “0-dimensional” (that is, isolated surfaces) and two “1-dimensional” families (surfaces parametrized by a complex number) with actions of large enough groups (order at least 8), which will be the protagonists of this chapter. The 0-dimensional families have the actions of C_{10} (the cyclic group of order 10), $C_3 \rtimes D_4$ (of order 24) and the already studied $GL_2(3)$ (of order 48), with signatures $(0; 10, 5, 2)$, $(0; 6, 4, 2)$ and $(0; 8, 3, 2)$ respectively. The 1-dimensional families have the actions of D_4 (the dihedral group of order 8) and D_6 (dihedral group of order 12), with signatures $(0; 4, 2, 2, 2)$ and $(0; 3, 2, 2, 2)$ respectively. The two 1-dimensional families share two common surfaces: the Bolza surface B and the surface A with the $C_3 \rtimes D_4$ -action.

Our strategy for finding lower bounds on the kissing number and upper bounds on the length of systoles for the $GL_2(3)$ -case (cf. chapter 2) it can also be applied directly to the surfaces with the group actions of $C_3 \rtimes D_4$ and C_{10} . In the section 3.1, we run the algorithm with them: first we construct a fundamental quadrilateral for the quotient sphere, then we use it to build a fundamental polygon for the surface where the group action is by permutations of the quadrilaterals, next we compute a homology basis of boundary curves and we find their geodesic representatives, and finally we performed the group action on the geodesic basis to obtain the systolic bounds by measuring the sizes of the orbits.

The first step must be adapted for 1-dimensional families, since the quotient surface has a hexagon as its fundamental polygon. In the subsection

3.2.1 of 3.2, we show how to construct such a hexagon, which depends on the choice of a complex number. The next three steps are analogous to the 0-dimensional case.

The lower bound on the kissing number depends only on the group action. Along the members of a 1-dimensional family, the *length of the systoles can change*. Unfortunately, we can compute this length explicitly only on a subfamily defined by a single real parameter. In this subfamily, we found the parameter value that maximizes the length of the systoles, which correspond precisely to the Bolza surface's bounds.

3.1 Zero-dimensional families

3.1.1 $C_3 \rtimes D_4$ -action

Let A be the surface with a $C_3 \rtimes D_4$ -action with signature $(0; 6, 4, 2)$. A presentation of the group with generators of order 6, 4, and 2 is

$$C_3 \rtimes D_4 \cong \langle x, y, z : x^6 = y^4 = z^2 = xyz = (xy^{-1})^2 = (x^2z^{-1})^2 = 1 \rangle. \quad (3.1)$$

In terms of these generators, the group has the following elements

$$C_3 \rtimes D_4 = \{x^j, x^j y, x^j xyx, x^j y^2 \mid j \in \{0, 1, 2, 3, 4, 5\}\}. \quad (3.2)$$

As we did in section 2.1 (step one), we determine a fundamental quadrilateral for the quotient sphere. It has sides of lengths $\rho = \cosh^{-1}(\sqrt{2})$ and $d = \cosh^{-1}(\sqrt{\frac{3}{2}})$ as depicted in figure 3.1,

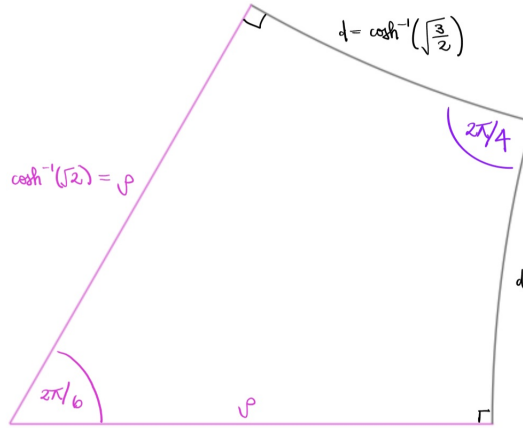


Figure 3.1: Fundamental quadrilateral for signature $(0; 6, 4, 2)$

and the fundamental polygon for the surface is formed by 24 copies of this quadrilateral, as in section 2.2 (step two). In 3.2 is shown the polygon with the side-pairing.

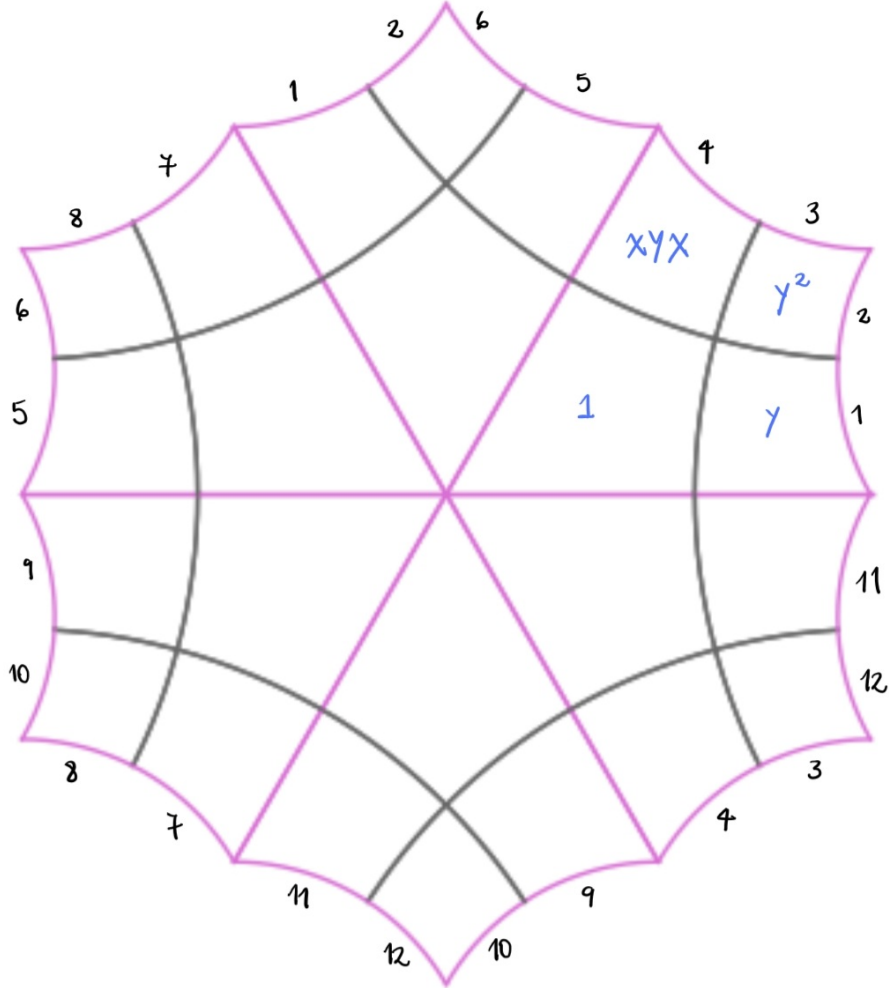


Figure 3.2: Fundamental polygon for A

A homology basis $\{\beta_1, \beta_2, \beta_3, \beta_4\}$ for the surface is depicted in figure 3.3 (step three); each basis curve has length $4 \cosh^{-1}(\sqrt{2})$. In the free homotopy class of β_1 lies the closed geodesic α_1 , with length $4d = 4 \cosh^{-1}(\sqrt{3/2})$.

Similarly, in the free homotopy class of β_2, β_3 and β_4 the curves α_2, α_3 and α_4 respectively are closed geodesics, each one with length $4d$, see figure 3.3.

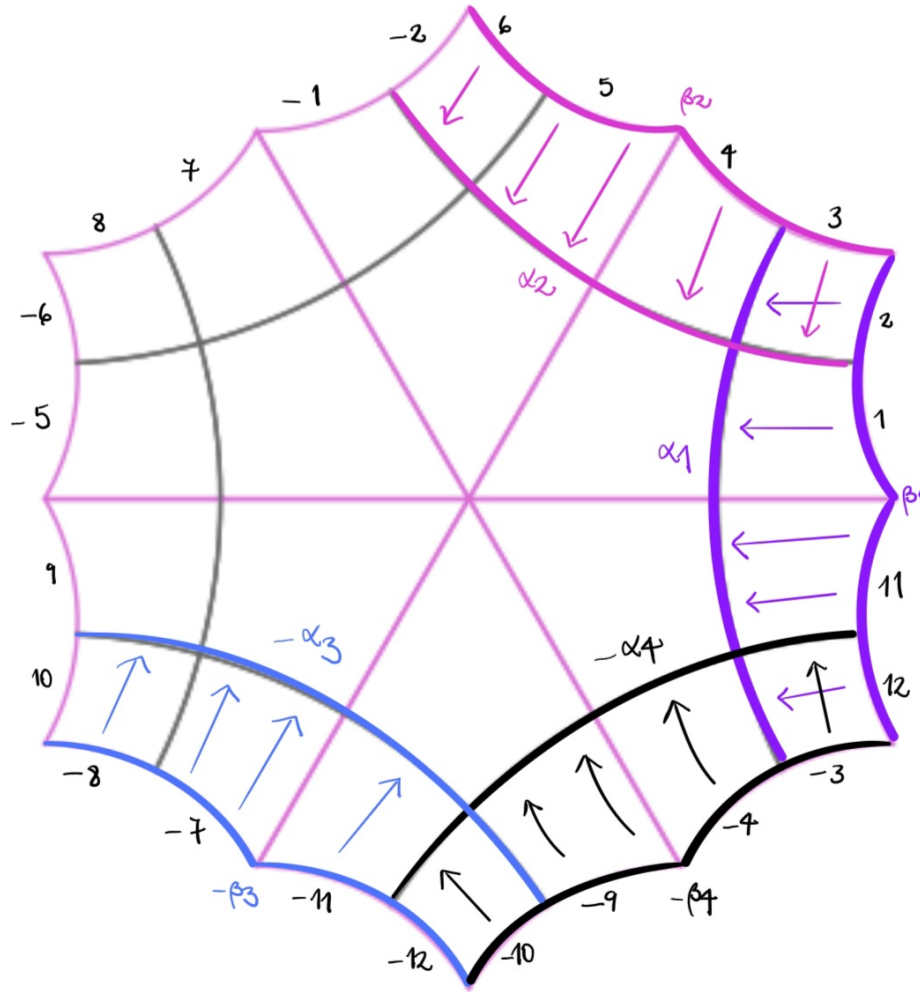


Figure 3.3: Geodesics on the free homotopy class of the basis curves

Following step 4 (section 2.4), the action of the generators x and y of $C_3 \rtimes D_4$ in the geodesic basis $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ of $H_1(A)$ is represented by the

following matrices.

$$H_1(x) = \begin{pmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad H_1(y) = \begin{pmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \quad (3.3)$$

The stabilizer subgroup of α_1 is $\text{Stab}(\alpha_1) = \{H_1(1), H_1(x^5y), H_1(y^2), H_1(yx)\}$, which has four elements. Thus, the orbit of α_1 has 6 elements.

All these computations leads us to the following result for the surface A .

Proposition 3.1. *The systolic bounds for the surface A with $C_3 \rtimes D_4$ -action are*

$$\text{sys}(A) \leq 4 \cosh^{-1} \left(\sqrt{\frac{3}{2}} \right) \approx 2.6339$$

$$\text{Kiss}(A) \geq 6.$$

Thus, the systoles of A are shorter than the Bolza surface's ones.

Notice that the bound on $\text{Kiss}(A)$ does not imply, a priori, that A has fewer systoles than B .

3.1.2 C_{10} -action

Let D be the surface with C_{10} -action with signature $(0; 10, 5, 2)$. A presentation of the group with generators of orders 10, 5, and 2 is the following.

$$C_{10} \cong \langle x, y, z \mid x^{10} = y^5 = z^2 = xzy = zxy = 1 \rangle. \quad (3.4)$$

In terms of these generators, the group consist of the elements

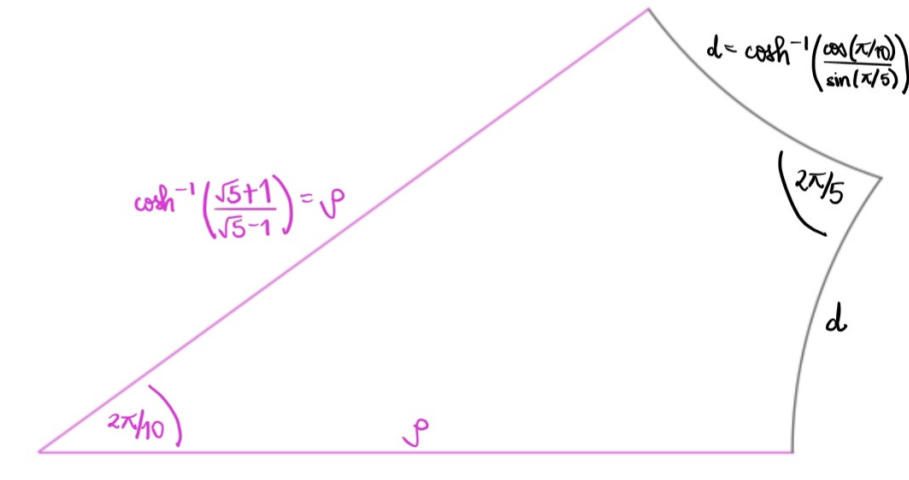
$$C_{10} = \{1, x, x^2, x^3, x^4, x^5, x^6, x^7, x^8, x^9\}. \quad (3.5)$$

Step one: The fundamental quadrilateral for the quotient sphere has

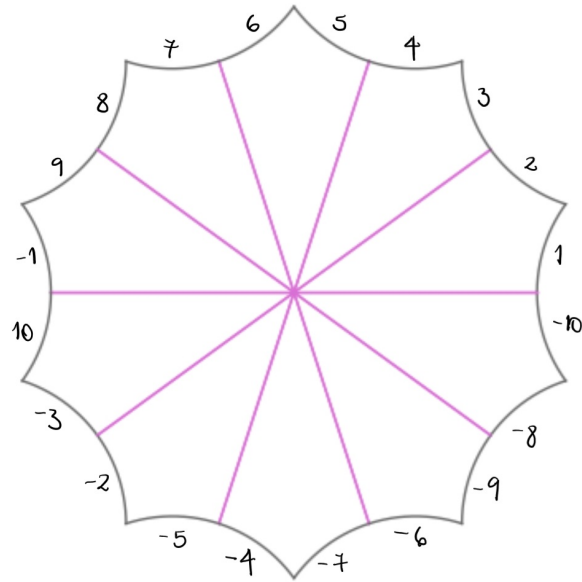
$$\text{sides } \rho = \cosh^{-1} \left(\frac{\cos(\pi/5)}{\sin(\pi/10)} \right) = \cosh^{-1} \left(\frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right)$$

$$\text{and } d = \cosh^{-1} \left(\frac{\cos(\pi/10)}{\sin(\pi/5)} \right) = \cosh^{-1} \left(\sqrt{\frac{10 + 2\sqrt{5}}{10 - 2\sqrt{5}}} \right), \text{ and it is depicted}$$

in figure 3.4.


 Figure 3.4: Fundamental quadrilateral for signature $(0; 10, 5, 2)$

Step two: A fundamental polygon for D is illustrated in figure 3.5. The polygon has 10 copies of the fundamental quadrilateral of figure 3.4.


 Figure 3.5: Fundamental polygon for D

Step three: A homology basis $\{\beta_1, \beta_2, \beta_3, \beta_4\}$ formed by boundary curves of the surface is shown in figure 3.6. Each basis curve has length $4d = 4 \cosh^{-1} \left(\frac{\cos(\pi/10)}{\sin(\pi/5)} \right)$. A geodesic representative α_1 in the homotopy class of β_1 is also depicted. As α_1 is the geodesic connecting the endpoints of β_1 , using `hyperbolic_length` we compute numerically its length: it is approximately 3.2338, a greater value than the Bolza surface's systolic length (compare with remark 2.4).

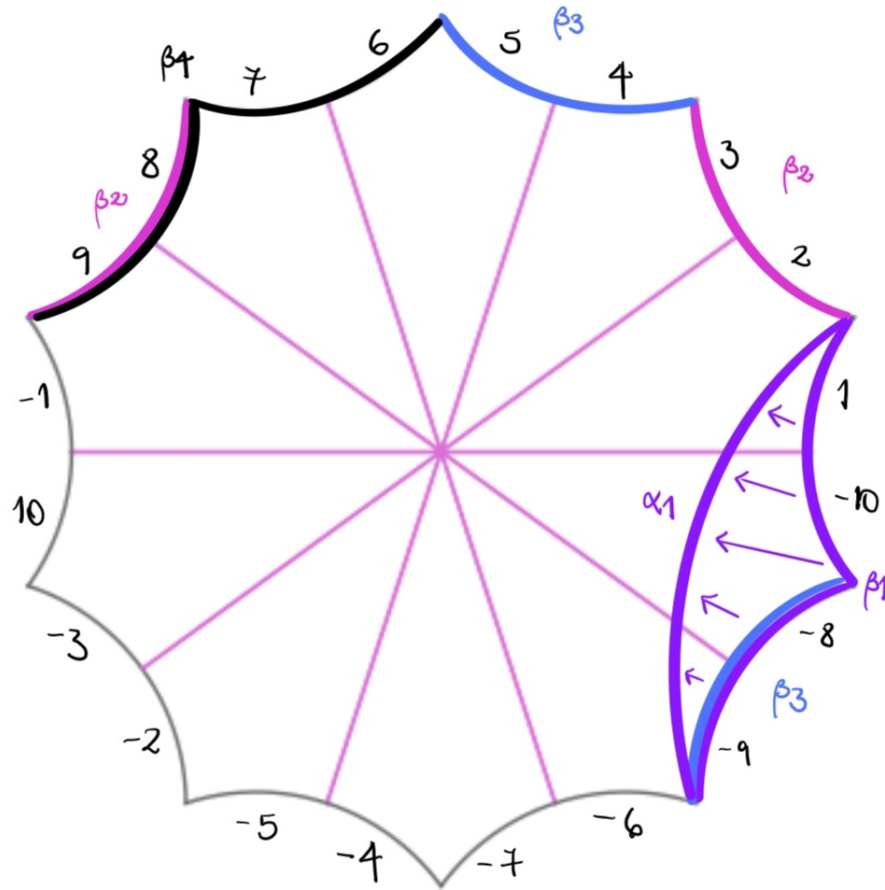


Figure 3.6: Homology basis for D . The pink curve β_2 and the blue β_3 seems to be discontinuous, but their endpoints are identified. The curve β_2 shares and edge with β_4 , and β_3 shares one with β_1 .

Step four: The action of the generators x and y on C_{10} on the basis $\{\beta_1, \beta_2, \beta_3, \beta_4\}$ is represented by the matrices

$$H_1(x) = \begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad H_1(y) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix} \quad (3.6)$$

The stabilizer subgroup of β_1 is $\text{Stab}(\beta_1) = \{1, H_1(x^5)\}$. Thus, the orbit of β_1 under the action of $H_1(D)$ has five elements.

Thus, all these computations leads us the following result for the surface D .

Proposition 3.2. *The systolic bounds for the surface D with C_{10} -actions are*

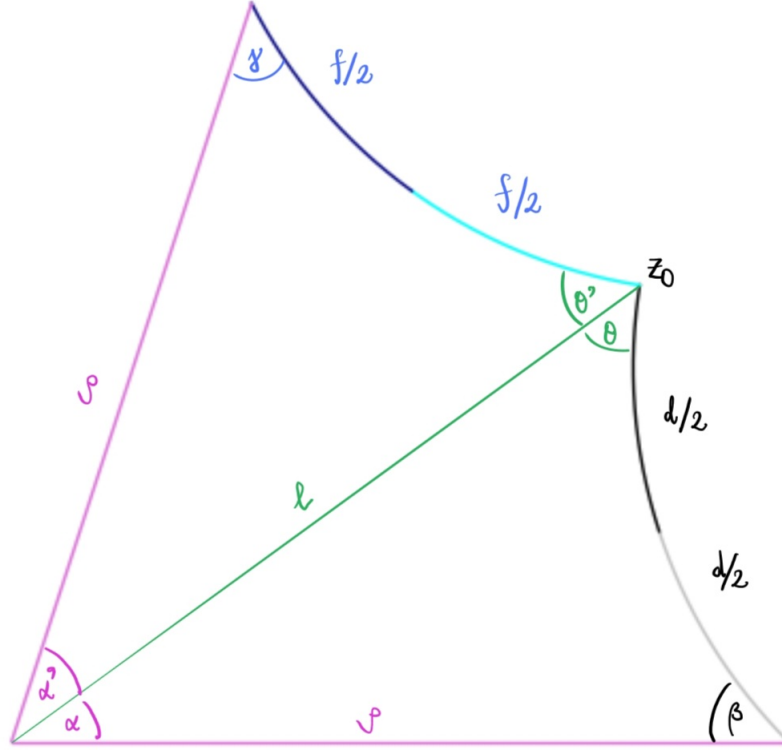
$$\begin{aligned} \text{sys}(D) &\leq 3.2338 \\ \text{Kiss}(D) &\geq 5. \end{aligned}$$

Observe that from these systolic bounds we can not conclude that D has longer systoles than B , or that it has fewer systoles than B .

3.2 One-dimensional families

3.2.1 Fundamental hexagons

In this section we focus on one-dimensional families on $\mathcal{M}_2$ whose members admit a group action with signature $(0; m, 2, 2, 2)$. According to [Behn et al., 2013, proposition 3.7], the fundamental polygon constructed in step one is now a hexagon depending on *two real parameters*. The general hexagon is depicted in figure 3.7. This polygon has sides of lengths $\rho, d/2, d/2, f/2, f/2$, and ρ ; its angles are $2\pi/m, \beta, \theta + \theta'$, and γ , subject to the equation $\beta + \theta + \theta' + \gamma = \pi$. The diagonal joining the origin to the vertex z_0 has length ℓ and divides the angle of $2\pi/m$ into two angles α and α' .

Figure 3.7: General hexagon for the signature $(0; m, 2, 2, 2)$.

According to [Behn et al., 2013, Proposition 3.7], the values ρ and $d/2$ parametrizes a hexagon. Due to the (heavy) trigonometric computations involved in the general case, we compute systolic bounds on a subfamily that depends on a *single* real parameter. We add the hypothesis $\alpha' = \alpha = \pi/m$, so the hexagon depends on the parameter θ (see figure 3.7 for the angle's labelling). The hexagons of this subfamily are depicted in figure 3.8.

By the Gauss-Bonnet theorem, a fundamental polygon for the action of a Fuchsian group Δ with signature $(0; m, 2, 2, 2)$ must be a *hexagon* with the sum of whose interior angles equal to $\frac{2\pi}{m} + 3\pi$. Such a Fuchsian group has the following presentation.

$$\Delta \cong \langle X, Y, Z, W \mid X^m = Y^2 = Z^2 = W^2 = XYZW = 1 \rangle \quad (3.7)$$

$$\cong \langle X, Y, Z \mid X^m = Y^2 = Z^2 = (XYZ)^2 = 1 \rangle \quad (3.8)$$

Figure 3.8 depicts a fundamental hexagon for a Δ -action on $\mathbb{D}$ with the desired signature (drawn using `quadrilateral_m22`, see 4.3). This

construction depends on a *real* parameter θ , an angle between 0 and π . The hexagon has two sides of length ρ and four of length $d/2$, with angles $\frac{2\pi}{m}, \frac{\pi}{2} - \frac{\theta}{2}, \pi, \theta, \pi$, and $\frac{\pi}{2} - \frac{\theta}{2}$. We denote by ℓ the distance between the vertex z_0 (whose angle is θ) and the origin, and by s the distance between the vertex z_1 (whose angle is π) and the origin. By symmetry, the distance between z_2 (the other vertex with angle π) and the origin is s . Observe that the hexagon appears to be a quadrilateral, since the angles at z_1 and z_2 are both π .

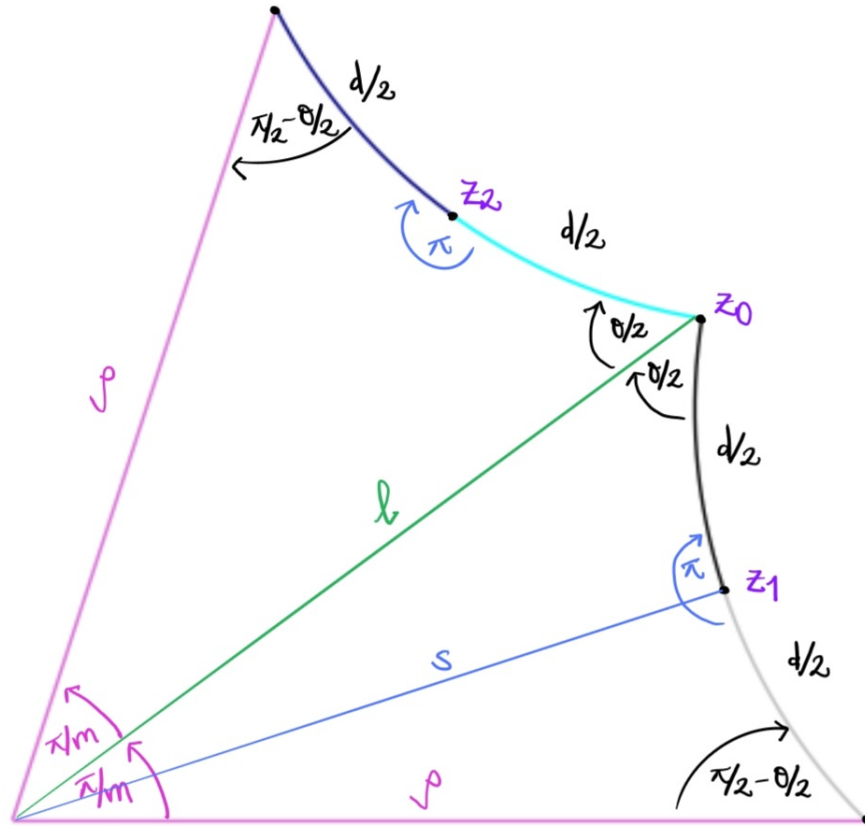


Figure 3.8: Fundamental hexagon for the real subfamily.

For the parameter θ , by the hyperbolic trigonometric laws, we obtain

the following formulas.

$$\cosh(d) = \frac{2 \cos(\pi/m)}{\sin(\theta)} + 1 \quad (3.9)$$

$$\cosh(d/2) = \sqrt{\frac{\cosh(d) + 1}{2}} = \sqrt{\frac{\cos(\pi/m)}{\sin(\theta)} + 1} \quad (3.10)$$

$$\cosh(\rho) = \cot(\pi/m) \tan(\theta/2) + \sin(\pi/m)^{-1} \quad (3.11)$$

$$\cosh(\ell) = \cot(\pi/m) \cot(\theta/2) + \sin(\pi/m)^{-1} \quad (3.12)$$

$$\cosh(s) = \frac{1}{\sin(\pi/m)} \sqrt{\frac{\cos(\pi/m)}{\sin(\theta)} + 1}. \quad (3.13)$$

As in section 2.1, we determine the Euclidean coordinates for the vertices of the hexagon. The real segment of hyperbolic length ρ has Euclidean length

$$r = \sqrt{\frac{\cos(\pi/m) \tan(\theta/2) - \sin(\pi/m) + 1}{\cos(\pi/m) \tan(\theta/2) + \sin(\pi/m) + 1}}. \quad (3.14)$$

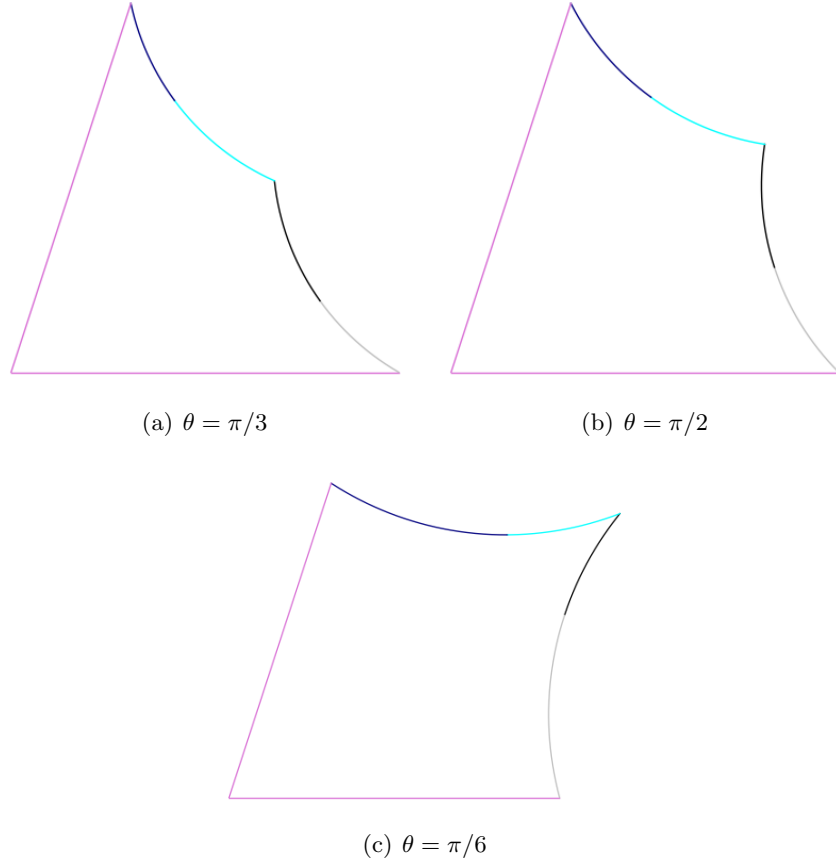
The vertex z_0 is the complex number $Re^{i\pi/m}$, with R the real number

$$R = \sqrt{\frac{\cos(\pi/m) \cot(\theta/2) - \sin(\pi/m) + 1}{\cos(\pi/m) \cot(\theta/2) + \sin(\pi/m) + 1}}. \quad (3.15)$$

The vertices z_1 and z_2 have modulus $t = \sqrt{\frac{\cosh(s) - 1}{\cosh(s) + 1}}$ and argument δ , with δ satisfying

$$\cot(\delta) = \frac{\tan(\theta/2) + \cos(\pi/m)}{\sin(\pi/m)}. \quad (3.16)$$

The figure 3.9 illustrates the effect of choosing three different values of θ , for a fixed m .

Figure 3.9: Different values of θ .

We conclude this section by showing the concrete generators of the Fuchsian group Δ . The generator X is an order m rotation centered at the origin, the generator Y is the order two rotation centered at z_1 , and Z is the order two rotation centered at z_2 .

$$X = \begin{pmatrix} e^{2i\pi/m} & 0 \\ 0 & 1 \end{pmatrix} \quad (3.17)$$

$$Y = \begin{pmatrix} -(1 + |z_1|^2) & 2z_1 \\ -2\bar{z}_1 & 1 + |z_1|^2 \end{pmatrix} \quad (3.18)$$

$$Z = \begin{pmatrix} -(1 + |z_2|^2) & 2z_2 \\ -2\bar{z}_2 & 1 + |z_2|^2 \end{pmatrix} \quad (3.19)$$

$$(3.20)$$

In the next subsections, we perform the steps two (section 2.2), three (section 2.3) and four (section 2.4) of our algorithm. First, we construct a fundamental polygon for the members of the subfamilies (step two), then we determine the side-pairing and a homology basis (step three), and finally we obtain a representation of the group action on that basis (step four) to compute a bound on the kissing number.

3.2.2 Family with D_6 -action

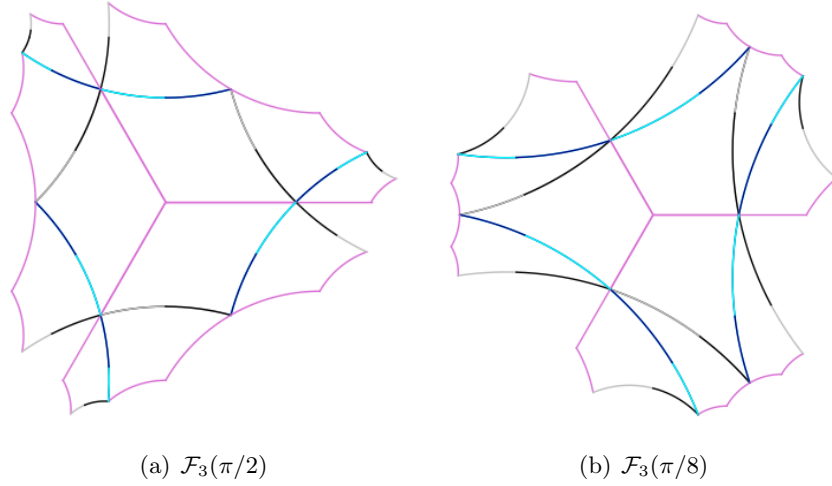
The dihedral group D_6 (of order 12) acts with signature $(0; 3, 2, 2, 2)$ in a real family $\mathcal{F}_3$ in $\mathcal{M}_2$. The member of the family corresponding to the value $\theta \in (0, \pi)$ will be denoted by $\mathcal{F}_3(\theta)$. The group has the following presentation with generators of order 3, 2, and 2.

$$D_6 \cong \langle x, y, z, w \mid x^3 = y^2 = z^2 = w^2 = zwx^{-1} = xyzw = zx^{-1}zx \rangle \quad (3.21)$$

In terms for these generators, the elements of the group are

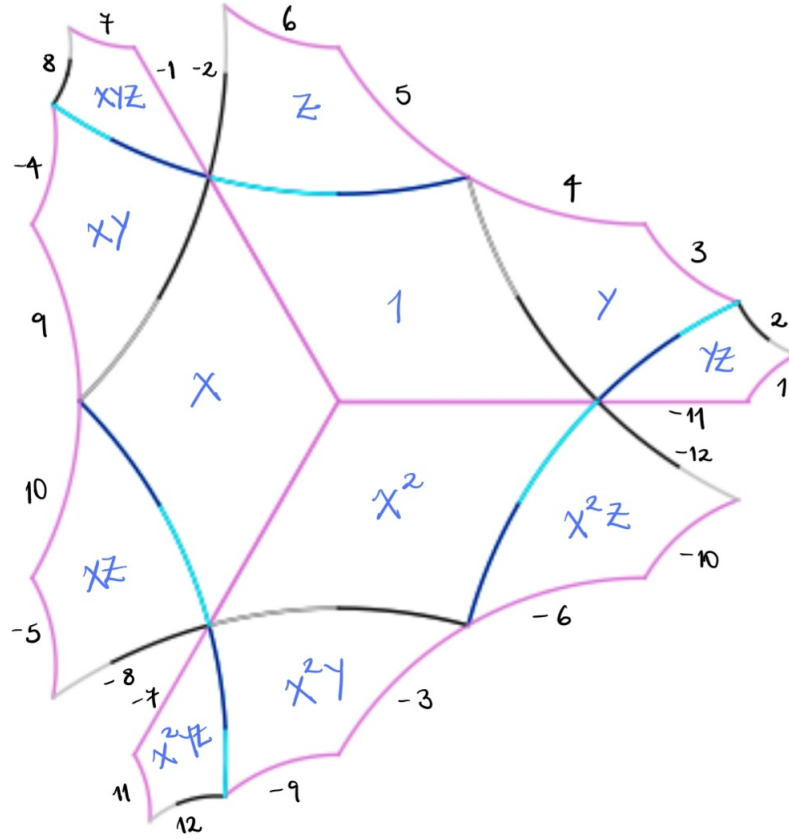
$$D_6 = \{1, y, x^2y, x, x^2, xy, yz, z, x^2z, xyz, x^2yz, xz\}. \quad (3.22)$$

The fundamental polygons for $\mathcal{F}_3(\pi/2)$ and $\mathcal{F}_3(\pi/8)$ are depicted in figure 3.10 (drawn with `FundamentalPolygon_m222`, see section 4.3).

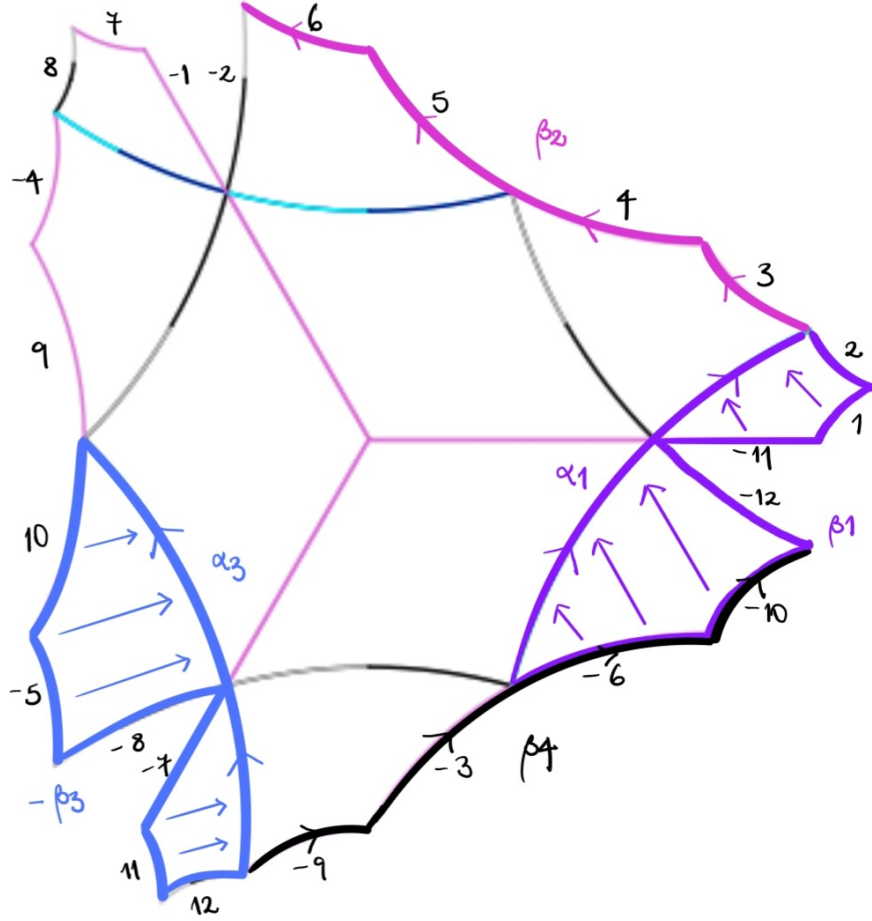
Figure 3.10: Some fundamental polygons on $\mathcal{F}_3$.

Remark 3.3. *Observe that the side-pairing depends only on the group. Therefore, the boundary curves forming a homology basis and the action of D_6 on that basis are the same for all members in $\mathcal{F}_3$. We illustrate these steps of our algorithm with the polygon $\mathcal{F}_3(\pi/2)$.*

The side-pairing in $\mathcal{F}_3$ is depicted in figure 3.11.

Figure 3.11: Side-pairing on $\mathcal{F}_3$

A homology basis $\{\beta_1, \beta_2, \beta_3, \beta_4\}$ is illustrated in figure (3.12). The curves α_1 and α_3 are geodesic representatives in the homotopy class of β_1 and β_3 respectively, and both have length $2d$ (d as in (3.9)). The curves β_2 and β_4 have length 4ρ (ρ as in (3.11)).


 Figure 3.12: Homology basis in $\mathcal{F}_3$.

The action of D_6 on the homology basis $\{\alpha_1, \beta_2, \alpha_3, \beta_4\}$ is represented by the matrices

$$H_1(x) = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (3.23)$$

$$H_1(y) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (3.24)$$

$$H_1(z) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (3.25)$$

The stabilizer subgroup of α_1 is trivial. This, the kissing number of $\mathcal{F}_3$ is at least 12.

Respect to the systolic length, the closed geodesic α_1 has length $2d = 2 \cosh^{-1} \left(\frac{2 \cos(\pi/3)}{\sin(\theta)} + 1 \right)$. By the derivative criterion, the value $2d$ attains its minimum at $\theta = \pi/2$. Specifically, with $\theta = \pi/2$ the curve has length $2 \cosh^{-1}(2) \approx 2.6339$. The surface $\mathcal{F}_3(\pi/2)$ is precisely the surface A , the one with the $C_3 \rtimes D_4$ (see subsection 3.1.1).

Thus, we have obtained the following result.

Proposition 3.4. *The systolic bounds for the real subfamily $\mathcal{F}_3$ with D_6 -action are*

$$\begin{aligned} \text{sys}(\mathcal{F}_3(\pi/2)) &\leq 2.6339 \\ \text{Kiss}(\mathcal{F}_3) &\geq 12. \end{aligned}$$

Notice that the other surfaces, parametrized by different values of θ , have bigger values as upper bounds on the systolic length. This does not imply that they actually have the longer systoles.

3.2.3 Family with D_4 -action

The dihedral group D_4 (of order 8) acts with signature $(0; 4, 2, 2, 2)$ in a real family $\mathcal{F}_4$. As in the previous section, we denote by $\mathcal{F}_4(\theta)$ the member of the family associated to the value $\theta \in (0, \pi)$. The group has the following presentation with generators of order 4, 2, and 2.

$$D_4 \cong \langle x, y, z \mid x^4 = y^2 = z^2 = (xyz)^2 = x^{-1}yz = 1 \rangle \quad (3.26)$$

Its elements are the following.

$$D_4 = \{1, y, x^3y, x, x^3, xy, x^2y, x^2\}. \quad (3.27)$$

Step two: The fundamental polygons for $\mathcal{F}_4(\pi/2)$ and $\mathcal{F}_4(\pi/8)$ are depicted in figure 3.13.

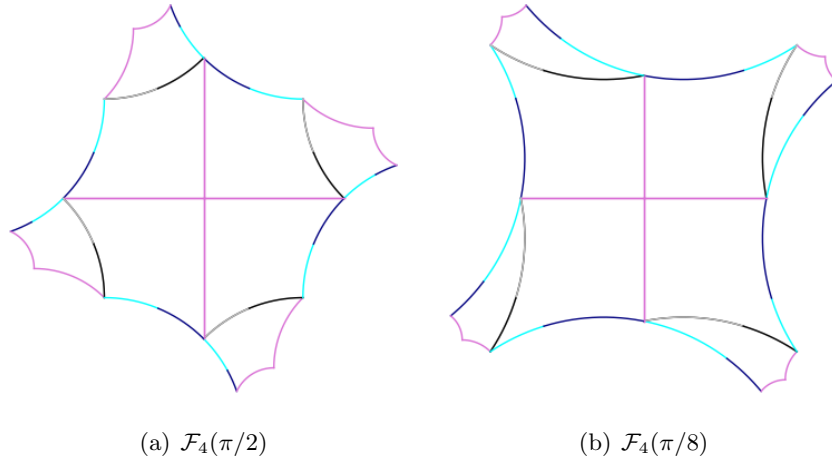
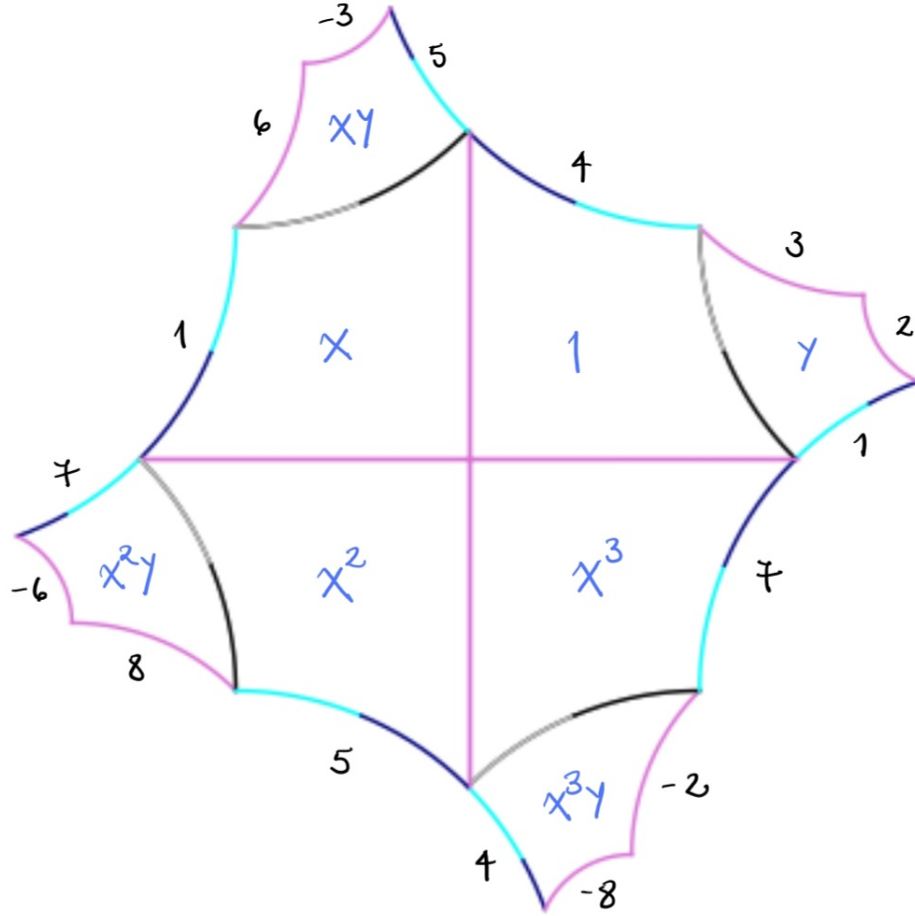
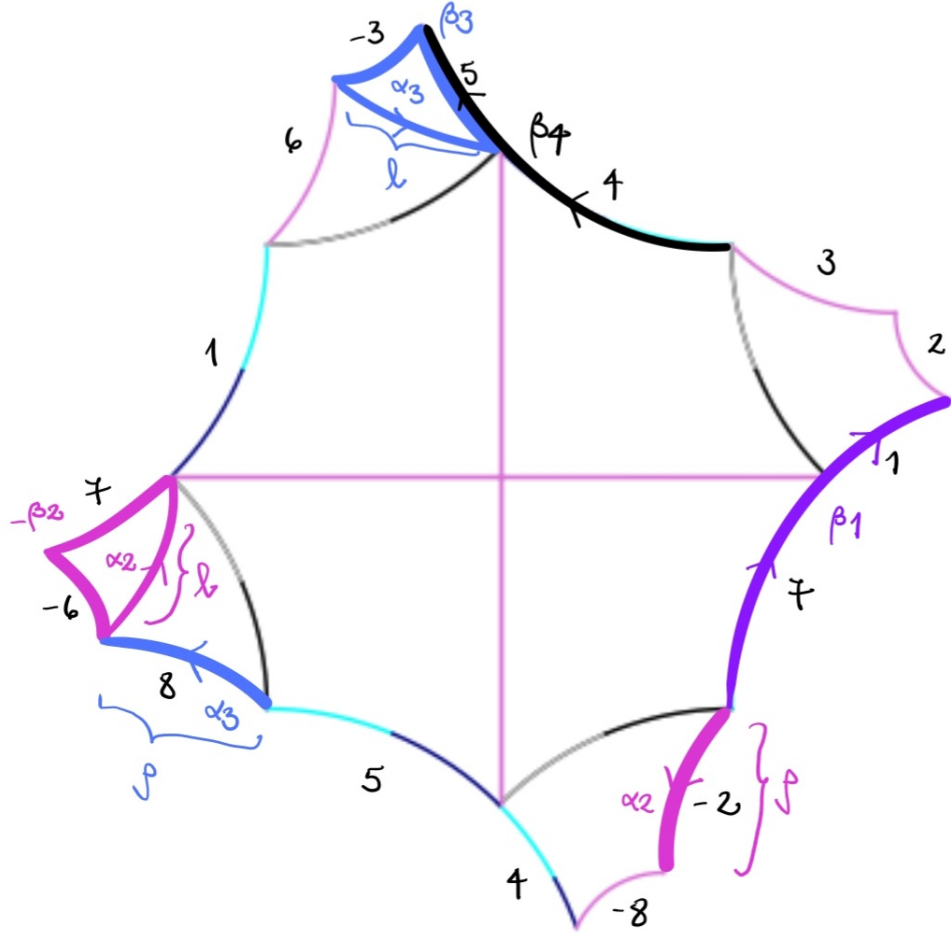


Figure 3.13: Some fundamental polygons in $\mathcal{F}_4$.

Step three: The side pairing in $\mathcal{F}_4$ is illustrated in figure 3.14.

Figure 3.14: Side-pairing in $\mathcal{F}_4$.

A homology basis $\{\beta_1, \beta_2, \beta_3, \beta_4\}$ is shown in figure 3.15. The curves β_1 and β_4 are closed geodesics, and both have length $2d$ (d as in (3.9)). The curves α_2 and α_3 are closed geodesic homotopic to β_2 and β_3 respectively, and both have length $\ell + \rho$ (ℓ as in (3.12) and ρ as in (3.11)).

Figure 3.15: Homology basis in $\mathcal{F}_4$.

Step four: The action of D_4 on the homology basis $\{\beta_1, \beta_2, \beta_3, \beta_4\}$ is represented by the following matrices.

$$H_1(x) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix} \quad (3.28)$$

$$H_1(y) = \begin{pmatrix} 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{pmatrix} \quad (3.29)$$

$$H_1(z) = \begin{pmatrix} -1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix}. \quad (3.30)$$

For the parameter θ , the curve β_1 has length $2d = 2 \cosh^{-1}(\sqrt{2} \sin(\theta)^{-1} + 1)$ and α_2 has length

$$\ell + \rho = \cosh^{-1}(\tan(\theta/2) + \sqrt{2}) + \cosh^{-1}(\cot(\theta/2) + \sqrt{2}). \quad (3.31)$$

Both quantities attain its minimum at $\pi/2$ and they coincide for that value. In fact, $2d = 2 \cosh^{-1}(\sqrt{2} + 1) = \ell + \rho$ for $\theta = \pi/2$. The surface $\mathcal{F}_4(\pi/2)$ has the same systolic length as the Bolza surface.

For the values of θ such that $2d \leq \ell + \rho$, since the stabilizer subgroup of β_1 is $\text{Stab}(\beta_1) = \{1, H_1(x^3y), H_1(xy), H_1(x^2)\}$, we obtain that $\text{Kiss}(\mathcal{F}_4(\theta)) \geq 3$. In the other case when $\ell + \rho \leq 2d$, as the stabilizer subgroup of β_2 is trivial we obtain the bound $\text{Kiss}(\mathcal{F}_4(\theta)) \geq 8$.

All these computations leads us to the following result.

Proposition 3.5. *The systolic bounds for the real subfamily with D_4 -action are*

$$\text{sys}(\mathcal{F}_4(\pi/2)) \leq 2 \cosh^{-1}(\sqrt{2} + 1) \quad (3.32)$$

$$\text{Kiss}(\mathcal{F}_4) \geq 8. \quad (3.33)$$

Observe that the other surfaces, parametrized by different values of θ , have bigger values as upper bounds on the systolic length. This does not imply that they actually have the longer systoles.

3.3 Conclusions

We confirm that the Bolza surface provides the best bound on the kissing number on $\mathcal{M}_2$. However, we can not conclude a priori that B has the longest systoles. The following table collects the values obtained from our algorithm.

GAP ID	Signature	Kiss lower bound	sys upper bound ($\approx$)	See in
(48,29)	(0;8,3,2)	12	3.0571	Rmk 2.4 , Thm 2.6
(24,8)	(0;6,4,2)	6	2.6339	Prop 3.1
(10,2)	(0;10,5,2)	5	3.2338	Prop 3.2
(12,4)	(0;3,2,2,2)	12	2.6339	Prop 3.4
(8,3)	(0;4,2,2,2)	8	3.0571	Prop 3.5

Chapter 4

Appendix

This chapter presents the complete Sage code, organized into three categories: hyperbolic geometry constructions, group manipulations, and fundamental polygons.

The first one, hyperbolic geometry constructions, includes drawing methods for hyperbolic geodesics and polygons. Next, in group manipulations, we determine a finitely presented GAP group with generators prescribed by a signature. We also compute a list of matrices that represent the group action on the fundamental polygon of the surface. Finally, in fundamental polygons, we can draw a fundamental quadrilateral for the quotient sphere, and a polygon for a surface endowed with a group action.

The formulas and constructions are based on [Anderson, 2006, Sections 5.3, 5.4 and 5.6], [Beardon, 2012, Chapter 7], and [Behn et al., 2013, sections 3, 4, and 5] We posted the code at the following GitHub repository <https://github.com/Cami-GV/Master-thesis.git>

4.1 Hyperbolic geometry constructions

```

z,w,vertex,c = var('z,w,vertex,c', domain='complex')
t,r,R,e = var('t,r,R,e', domain='real')
C = circle((0,0), 1, rgbcolor=(0,0,0))

Id = Matrix(CC,[[1,0],[0,1]])

def apply_mobius(M: matrix, z:complex) -> complex:
    """
    Input:
    -M: 2x2 complex matrix, representing a Möbius transformation
    -z: Complex number in the unit disc

    Output:
    -Mz: Image of z under M

    Example:
        if M=[[a,b],[c,d]], apply_mobius(M,z)=(az+b)/cz+d
    """
    Mz = (M[0][0]*z+M[0][1])/(M[1][0]*z+M[1][1])
    return Mz

def invert_mobius(M:matrix) -> matrix:
    """
    Return a matrix N, representing the inverse Möbius
    transformation of M
    """
    N = [[M[1][1],-M[0][1]],[-M[1][0],M[0][0]]]
    return N

```

```

def apply_translation(c:complex , z:complex) -> complex:
    '''
    A hyperbolic translation is a Möbius transformation of the form
     $t \rightarrow (t-c)/(1-t\overline{c})$ 

    Input:
    -c: Complex number. Represents the preimage of 0
    -z: Complex number

    Output:
    -Mz: Image of z under a hyperbolic translation sending c to 0
    '''
    M = [[1,-c],[-c.conjugate(),1]]
    Mz = apply_mobius(M,z)
    return Mz

```



```

def hyperbolic_length(z: complex, w:complex) -> float :
    '''
    Return the hyperbolic distance between z and w

    Procedure:
    1. Map z to T(z)=0 and w to T(w) by a hyperbolic translation
       (isometry of the unit disc)
    2. Compute the distance between 0 and T(w)
    '''
    a = apply_translation(z,w)
    l = log((1+a.abs()/(1-a.abs())))
    return float(l)

```

```

def hyperbolic_angle(z: complex, w: complex, vertex: complex) -> float:
    """
    Return the angle between the vectors z - vertex and w - vertex

    Procedure:
    1. Map vertex to T(vertex)=0 with a hyperbolic translation
       (isometry of the unit disc)
    2. Compute the angle between T(z) and T(w)
    """
    Tz = apply_translation(vertex, z)
    Tw = apply_translation(vertex, w)
    angle = arccos((Tz.real()*Tw.real()+Tz.imag()*Tw.imag())/(Tz.abs()*Tw.abs()))
    return angle

```



```

def hyperbolic_geodesic(z: complex, w: complex, color=[]) -> plot:
    """
    Input:
    -z: Complex number. It is the starting point of the geodesic
    -w: Complex number. It is the ending point of the geodesic
    -color:

    Output:
    -P: Graphic object representing the hyperbolic geodesic from z to w

    Method:
        The geodesic starting at z and ending at w is the curve
         $t \rightarrow (tT(w)+z)/(1+t\bar{z}T(w))$ ,
        T is Möbius transformation sending z to 0
    """
    T = apply_translation(z, w)
    P = parametric_plot([apply_translation(-z, t*T).real(), \
        apply_translation(-z, t*T).imag()], \
        (t, 0, 1), rgbcolor=color, axes=false)
    return P

```

```

def draw_Polygon(points:list, colors=[]) -> plot:
    '''
    Input:
    -points: List of complex numbers, representing the vertices of a polygon
    -colors: List of strings for color codes.
        By default, all edges are blue

    Output:
    -poly: Graphic object, representing a hyperbolic polygon with given vertices
    '''
    if colors==[]:
        colors = [[0,0,1] for k in range(len(points))]
    edges = [hyperbolic_geodesic(points[k],points[(k+1)%len(points)], colors[k]) \
        for k in range(len(points))]
    poly = sum(edges)
    return poly

```



```

def rotate_Polygon(vertices: list, rotation: matrix) -> list:
    '''
    Input:
    -vertices: List of complex numbers, representing the vertices of a polygon
    -rotation: Matrix, representing a Möbius transformation

    Output:
    -rotated_vertices: List of complex numbers.
        Contains the images of vertices under the transformation rotation
    '''
    rotated_vertices = [apply_mobius(rotation,vertices[k]) \
        for k in range(len(vertices))]
    return rotated_vertices

```

4.2 Group manipulations

```

from io import StringIO

def search_generators(G,Sig:list) -> list:
    '''
    Input:
    -G: Finitely presented GAP group
    -Sig: List of the form [m,n,2], which comes from a signature
          of type (0;m,n,2)

    Output:
    -search: List of lists. The i-th list contains all elements of G
             of order Sig[i].
    '''
    group_elements = libgap.Elements(G)
    search = []
    for m in range(len(Sig)):
        could_be = []
        for k in range(len(group_elements)):
            if libgap.Order(group_elements[k])==Sig[m]:
                could_be.append(group_elements[k])
        search.append(could_be)
    return search

```

```

generators = []
def potencial_generators(G,search:list, acumulados:list=[]) -> list:
    '''
    Input:
    -G: Finitely presented GAP group
    -search: List obtained from search_generators. Contains elements of G
            with orders prescribed by a signature
    -acumulados: variable list. By default is the empty list

    Output:
    -generators: List of all possible sets of generators of G
                with the prescribed orders
    '''
    if len(search)==0:
        if prod(acumulados) == libgap.One(G):
            S = libgap.Subgroup(G, acumulados)
            if libgap.Index(G,S)==1:
                generators.append(acumulados)
    else:
        for g in search[0]:
            potencial_generators(G, search[1:], acumulados+[g])
    return generators

def with_signature_generators(id, Sig:list):
    '''
    Input:
    -id: GAP catalogue ID of a finite group
    -Sig: List of the form [m,n,2], which comes from a signature of type (0;m,n,2)

    Output:
    -K: Finitely presented GAP group isomorphic to the given ID,
        and with prescribed generators
    '''
    G = fp_representation(id)
    generators.clear()
    search = search_generators(G,Sig)
    gens = potencial_generators(G,search)
    K = libgap.IsomorphismFpGroupByGenerators(G,gens[0]).Image()
    return K

```

```

def as_matrices_mn2(K,Sig) -> list:
    '''
    Input:
    -K: Finitely presented GAP group, obtained using With_signature_generators
    -Sig: List of the form [m,n,2], which comes from
          a signature of type (0;m,n,2)

    Output:
    -elements_as_mat: List containing matrices.
                      These matrices represent the elements of K as Möbius transformations
    '''
    s = StringIO()
    elements = libgap.Elements(K)
    print(elements,file=s)
    rotations = rotations_mn2(Sig)
    X = rotations[0]
    Y = rotations[1]
    Z = rotations[0]*rotations[1]
    elements_str = [f.strip() for f in s.getvalue()[1:-2].replace('F1', 'X') \
                    .replace('F2', 'Y').replace('F3','Z').replace('^', '**').split(',')]
    elements_as_mat = [Id]+[sage_eval(expr, locals = {'X':X, 'Y':Y, 'Z':Z}) \
                          for expr in elements_str[1:]]
    return elements_as_mat

```

```

def as_matrices_m222(K,Sig, theta) -> list:
    '''
    Input:
    -K: Finitely presented GAP group, obtained using With_signature_generators
    -Sig: List of the form [m,2,2,2], which comes from
          a signature of type (0;m,2,2,2)

    Output:
    -elements_as_mat: List containing matrices.
                      These matrices represent the elements of K as Möbius transformations
    '''
    s = StringIO()
    elements = libgap.Elements(K)
    print(elements,file=s)
    rotations = rotations_m222(Sig[0],theta)
    X = rotations[0]
    Y = rotations[1]
    Z = rotations[2]
    W = rotations[3]
    elements_str = [f.strip() for f in s.getvalue()[1:-2].replace('F1', 'X') /
                    .replace('F2', 'Y').replace('F3','Z').replace('F4','W')/
                    .replace('^', '**').split(',')]]
    elements_as_mat = [Id]+[sage_eval(expr, locals = {'X':X, 'Y':Y, 'Z':Z, 'W':W})/
                          for expr in elements_str[1:]]
    return elements_as_mat

def as_word_in_G(K,elem):
    '''
    Input:
    -K: Finitely presented GAP group
    -elem: Expression in the generators of K.

    Output:
    - An element of K, equivalent to elem under the relations of the group
    '''
    element = libgap.Elements(K)
    factor = libgap.Factorization(K,elem) #expression in the generators of K
    for k in range(len(element)):
        if libgap.Factorization(K,element[k])==factor:
            return element[k]

```

```

def action_of_G_mn2(K,actions) -> list:
    """
    Input:
    -K: Finitely presented GAP group
    -actions: Two-matrices list.
           Contains the representing matrices of the action of the generators of K

    Output:
    -actions_as_mat: List of matrices.
           Contains the representing matrices for the actions of the elements of K

    """
    w= StringIO()
    elements = libgap.Elements(K)
    print(elements,file=w)
    S = actions[0]
    T = actions[1]
    U = S*T
    actions_str = [f.strip() for f in w.getvalue()[1:-2].replace('F1', 'S') \
                    .replace('F2', 'T').replace('F3','U').replace('^', '**').split(',')]
    actions_as_mat = [Matrix([[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1]])+ \
                       [sage_eval(expr, locals = {'S':S, 'T':T, 'U':U}) \
                        for expr in actions_str[1:]]
    return actions_as_mat

```

```

def stabilizers_mn2(K,vec,actions) -> list:
    '''
    Input:
    -K: Finitely presented GAP group
    -vec: Four-integer vector
    -actions: List of matrices.
           Constains the matrices representing the action of the generators of K
           on a given homology basis

    Output:
    -stab_elements: List of matrices.
           Contain the elements of K fixing vec under its corresponding action
    '''
    elem = libgap.Elements(K)
    stabs = []
    Actions = action_of_G_mn2(K,actions)
    for k in range(len(Actions)):
        if Actions[k]*vec==vec or Actions[k]*vec==-vec:
            stabs.append([Actions[k],k])
    print(len(stabs))
    stab_elements = []
    for m in range(len(stabs)):
        M = stabs[m]
        stab_elements.append(elem[M[1]])
    return stab_elements

```

```

def action_of_G_m222(K,actions) -> list:
    """
    Input:
    -K: Finitely presented GAP group
    -actions: Two-matrices list.
        Contains the representing matrices of the action of the generators of K

    Output:
    -actions_as_mat: List of matrices.
        Contains the representing matrices for the actions of the elements of K

    """
    w= StringIO()
    elements = libgap.Elements(K)
    print(elements,file=w)
    S = actions[0]
    T = actions[1]
    U = actions[2]
    V = S*T*U
    actions_str = [f.strip() for f in w.getvalue()[1:-2].replace('F1', 'S') /
        .replace('F2', 'T').replace('F3','U').replace('F4','V') /
        .replace('^', '**').split(',')]
    actions_as_mat = [Matrix([[1,0,0,0],[0,1,0,0],[0,0,1,0],[0,0,0,1]]) /
        +[sage_eval(expr, locals = {'S':S, 'T':T, 'U':U, 'V':V}) /
        for expr in actions_str[1:]]
    return actions_as_mat

```

```

def stabilizers_m222(K,vec,actions) -> list:
    '''
    Input:
    -K: Finitely presented GAP group
    -vec: Four-integer vector
    -actions: List of matrices.
            Constains the matrices representing the action of the generators of K
            on a given homology basis

    Output:
    -stab_elements: List of matrices.
            Contain the elements of K fixing vec under its corresponding action
    '''
    elem = libgap.Elements(K)
    stabs = []
    Actions = action_of_G_m222(K,actions)
    for k in range(len(Actions)):
        if Actions[k]*vec==vec or Actions[k]*vec==vec*-1:
            stabs.append([Actions[k],k])
    print(len(stabs))
    stab_elements = []
    for m in range(len(stabs)):
        M = stabs[m]
        stab_elements.append(elem[M[1]])
    return stab_elements

```

4.3 Fundamental polygons

```
def vertices_Polygon_mn2(Sig: list) -> list:
    '''
    Input:
    -Sig: List of the form [m,n,2], which comes from a signature
          of type (0;m,n,2)

    Output:
    -vertices: Four-complex numbers list, representing the vertices of
              a fundamental quadrilateral for the action with given signature

    Description:
        The formulas are derived from the hyperbolic trigonometric laws.
        The fundamental quadrilateral has angles  $2\pi/m$ ,  $\pi/2$ ,  $2\pi/n$ , and  $\pi/2$ .
    '''
    r = sqrt((cos(pi*Sig[1]^(-1))-sin(pi*Sig[0]^(-1)))/(cos(pi*Sig[1]^(-1)) \
        +sin(pi*Sig[0]^(-1))))).n()
    R = sqrt(cos(pi*(Sig[0]^(-1)+Sig[1]^(-1)))/cos(pi*(Sig[1]^(-1)-Sig[0]^(-1))))
    R = R.n()
    z_0 = CDF(R*cos(pi*Sig[0]^(-1)),R*sin(pi*Sig[0]^(-1)))
    z_1 = CDF(r*cos(2*pi*Sig[0]^(-1)),r*sin(2*pi*Sig[0]^(-1)))
    vertices = [0,CDF(r,0),z_0,z_1]
    return vertices
```

```

def draw_Polygon_mn2(Sig: list,color=[]) -> list:
    '''
    Input:
    -Sig: List of the form [m,n,2], which comes from a signature
          of type (0;m,n,2)
    -color: Four-string list for color codes.
            By default the colors are orchid-gray-gray-orchid

    Output:
    -P: Graphic object. Rrepresents a fundamental quadrilateral for the action
        on the unit disc with given signature.
    '''
    if len(color)==0:
        color = ['#DA70D6','#696969','#696969','#DA70D6']
    vertices = vertices_Polygon_mn2(Sig)
    P = draw_Polygon(vertices, color)
    return P

def rotations_mn2(Sig: list) -> list:
    '''
    Input:
    -Sig: List of the form [m,n,2], which comes from a signature
          of type (0;m,n,2)

    Output:
    - Two-matrices list, representing the generators of the uniformizing
      Fuchsian group for the quotient sphere, obtained under the action
      with given signature

    Description:
      The matrices are rotations of order m and n, centered at 0 and z_0
      respectively (see vertices_Polygon_mn2)
    '''
    verts = vertices_Polygon_mn2(Sig) #relevant vertices: 0 and 2
    angles = [CDF(cos(2*pi*(Sig[k]^(-1))),sin(2*pi*(Sig[k]^(-1)))) \
              for k in range(len(Sig))]
    A = Matrix(CC,[[angles[0],0],[0,1]])
    B = Matrix(CC,[[angles[1]-(verts[2].abs()^2), verts[2]*(1-angles[1])],
                  [-verts[2].conjugate()*(1-angles[1]), 1-(angles[1]*(verts[2].abs()^2))])
    return [A,B]

```

```

def fundamentalpolygon_mn2(id,Sig,colors=[]) -> plot:
    '''
    Input:
    -id: GAP catalogue ID of a finite group
    -Sig: List of the form [m,n,2], which comes
          from a signature of type (0;m,n,2)
    -colors: Four-strings list for color codes.
             By default the colors are orchid-gray-gray-orchid

    Output: Graphic object, representing a fundamental polygon for an action
             of the group with given ID and signature in the unit disc.
    '''
    if len(colors) == 0:
        colors = ['#DA70D6','#696969','#696969','#DA70D6']
    K = with_signature_generators(id,Sig)
    matrices = as_matrices(K,Sig)
    vertices = vertices_Polygon_mn2(Sig)
    rotated_vertices = [rotate_Polygon(vertices,matrices[k]) \
                        for k in range(len(matrices))]
    rotated_pieces = [draw_Polygon(rotated_vertices[k], colors) \
                      for k in range(len(matrices))]
    poly = sum(rotated_pieces)
    return poly

```

```

def vertices_m222(m: int, theta) -> list:
    '''
    Input:
    -m: Integer greater than 2
    -theta: an angle between 0 and pi,
            representing an element in the real-parametrized family of the action.

    Output:
    -vertices: Six-complex numbers list, representing the vertices of a fundamental
               hexagon for the action with signature (0;m,2,2,2).
               Two of the vertices are degenerate.

    Description:
        The formulas are derived from the hyperbolic trigonometric laws.
        The angles of the obtained polygon sum  $2\pi/m + 3\pi$ .
    '''
    tau = pi/m
    r = sqrt((cos(tau)*tan(theta/2)-sin(tau)+1)/(cos(tau)*tan(theta/2)+sin(tau)+1)).n()
    R = sqrt((cos(tau)*cot(theta/2)-sin(tau)+1)/(cos(tau)*cot(theta/2)+sin(tau)+1)).n()
    cosht = sqrt((cos(tau)/sin(theta))+1)*sin(tau)^(-1)
    s = sqrt((cosht-1)/((cosht+1))).n()
    delta = acot((tan(theta/2)+cos(tau))/sin(tau))
    vertices = [0, r,
                CDF(s*cos(delta),s*sin(delta)),
                CDF(R*cos(tau),R*sin(tau)),
                CDF(s*cos(2*tau-delta),s*sin(2*tau - delta)),
                CDF(r*cos(2*tau),r*sin(2*tau))]
    return vertices

```

```

def hexagon_m222(m: int, theta, colors=[]) -> plot:
    '''
    Input:
    -m: Integer, greater than 2
    -theta: an angle between 0 and pi,
            representing an element in the real-parametrized family of the action
    -colors: Six-strings for color codes.
            By default the colors are orchid-silver-black-aqua-navy-orchid

    Output:
    -quad: Graphic object representing a fundamental polygon for an action
            with signature (0;m,2,2,2).
    '''
    if len(colors)==0:
        colors = ['orchid', 'silver', 'black', 'aqua', 'navy','orchid']
    verts = vertices_m222(m, theta)
    hex = draw_Polygon(verts, colors)
    return hex

def rotations_m222(m: int, theta) -> list:
    '''
    Input:
    -m: Integer, grater than 2
    -theta: an angle between o and pi,
            representing and element in the real-parametrized family of the action.

    Output: Four-matrices list, representing the generators of a Fuchsian group
            acting with signature (0;m,2,2,2)
    '''
    verts = vertices_m222(m,theta)
    X = Matrix(CC,[[CDF(cos(2*pi/m),sin(2*pi/m)),0],[0,1]]) #order m
    Y = Matrix(CC,[[-(1+verts[2].abs()^2), 2*verts[2]],
                   [-2*verts[2].conjugate(), 1+verts[2].abs()^2]]) #order 2
    Z = Matrix(CC,[[-(1+verts[4].abs()^2), 2*verts[4]],
                   [-2*verts[4].conjugate(), 1+verts[4].abs()^2]]) #order 2
    W = X*Y*Z
    return [X,Y,Z,W]

```

```

def fundamental_polygon_m222(id, Sig, theta, colors=[]) -> plot:
    '''
    Input:
    -id: GAP catalogue ID for a finite group
    -Sig: List of the form [m,2,2,2], which comes from a signature
        of type (0;m,2,2,2)
    -theta: Angle between 0 and pi, representing a member for the
        real-parametrized family with the given action
    -colors: Six-strings for color codes.
        By default the colors are orchid-silver-black-aqua-navy-orchid

    Output: Graphic object, representing a fundamental polygon for an action
        of the group with given ID and signature in the unit disc.
    '''
    if len(colors)==0:
        colors = ['orchid', 'silver', 'black', 'aqua', 'navy','orchid']
    K = with_signature_generators(id,Sig)
    matrices = as_matrices_m222(K,Sig,theta)
    verts = vertices_m222(Sig[0], theta)
    rotated_vertices = [rotate_Polygon(verts,matrices[k]) /
        for k in range(len(matrices))]
    rotated_pieces = [draw_Polygon(rotated_vertices[k],colors) /
        for k in range(len(rotated_vertices))]
    poly = sum(rotated_pieces)
    return poly

```

4.4 Comments and limitations

The GAP's interface in Sage allows us to perform some group-theoretic manipulations, but their outputs need to be interpreted as Sage (or Python) objects to interact with graphics interfaces. For instance, the functions `with_signature_generators` (4.2) and `as_matrices_mn2` (4.2) require the elements of a given GAP group at the point of use.

These redundancies may impact efficiency with larger groups. However, we did not perform a thorough analysis of the algorithm's efficiency.

The steps for identifying boundary elements of the fundamental polygon are computed manually, as we are unable to fully implement them. Particularly, we need to identify the elements of the group that gives boundary curves to perform the side-pairing, to label and store properly the edges and vertices for homological computations, and to obtain the representative matrices of the group action on the basis curves.

Our functions are suitable for triangular signatures and the ones of the form $(0; m, 2, 2, 2)$ for $m \geq 3$, but some of them can be adapted for other types of signatures. For example, `as_matrices_mn2` (4.2) nests as many `replace` methods as generators of the given Fuchsian group, and the same phenomenon affects `action_of_G_mn2` (4.2). However, the function `vertices_Polygon_mn2` (4.3) comes from an explicit parametrization of the vertices of a fundamental polygon for the quotient surface, while `rotations_mn2` (4.3) depends on a concrete realization of the desired Fuchsian group as a group of Möbius transformations.

Finally, we strongly support our choice of a generating vectors, using `with_signature_generators` (4.2), in the uniqueness of topologically equivalent actions. Otherwise, it is necessary to develop and implement algorithms to distinguish non-equivalent actions of a given group.

Bibliography

- [Anderson, 2006] Anderson, J. W. (2006). *Hyperbolic geometry*. Springer Science & Business Media.
- [Beardon, 2012] Beardon, A. F. (2012). *The geometry of discrete groups*, volume 91. Springer Science & Business Media.
- [Behn et al., 2013] Behn, A., Rodríguez, R. E., and Rojas, A. M. (2013). Adapted hyperbolic polygons and symplectic representations for group actions on riemann surfaces. *Journal of Pure and Applied Algebra*, 217(3):409–426.
- [Broughton, 1991] Broughton, A. S. (1991). Classifying finite group actions on surfaces of low genus. *Journal of Pure and Applied Algebra*, 69(3):233–270.
- [Bustamante and Kordaß, 2017] Bustamante, M. and Kordaß, J.-B. (2017). Spaces of riemannian metrics. *Snapshots of Modern Mathematics from Oberwolfach*.
- [Carvacho Bustamante, 2010] Carvacho Bustamante, M. P. (2010). *Equivalence of Group Actions on Riemann Surfaces*. PhD thesis, Universidad de Chile, Santiago, Chile.
- [GAP-Group, 2019] GAP-Group (2019). Gap-groups, algorithms, and programming, version 4.10.2; 2019.
- [Greenberg and Harper, 2018] Greenberg, M. J. and Harper, J. (2018). *Algebraic topology: a first course*. CRC Press.
- [Jost, 2017] Jost, J. (2017). *Riemannian geometry and geometric analysis*. Springer, seventh edition.
- [Lee, 2003] Lee, J. M. (2003). Smooth manifolds. In *Introduction to smooth manifolds*. Springer.

- [Lee, 2018] Lee, J. M. (2018). *Introduction to Riemannian manifolds*. Springer, second edition.
- [Parlier, 2013] Parlier, H. (2013). Kissing numbers for surfaces. *Journal of Topology*, 6(3):777–791.
- [Rodriguez and Rojas, 2024] Rodriguez, R. E. and Rojas, A. M. (2024). A fruitful interaction between algebra, geometry, and topology: Varieties through the lens of group actions. *Notices of the American Mathematical Society*, 71(6).
- [Schmutz, 1993] Schmutz, P. (1993). Riemann surfaces with shortest geodesic of maximal length. *Geometric & Functional Analysis GAFA*, 3(6):564–631.
- [Schmutz-Schaller, 1998] Schmutz-Schaller, P. (1998). Geometry of riemann surfaces based on closed geodesics. *Bulletin of the American Mathematical Society*, 35(3):193–214.
- [Stein et al., 2020] Stein, W. et al. (2020). *Sage Mathematics Software (Version 9.0)*. The Sage Development Team. <http://www.sagemath.org>.